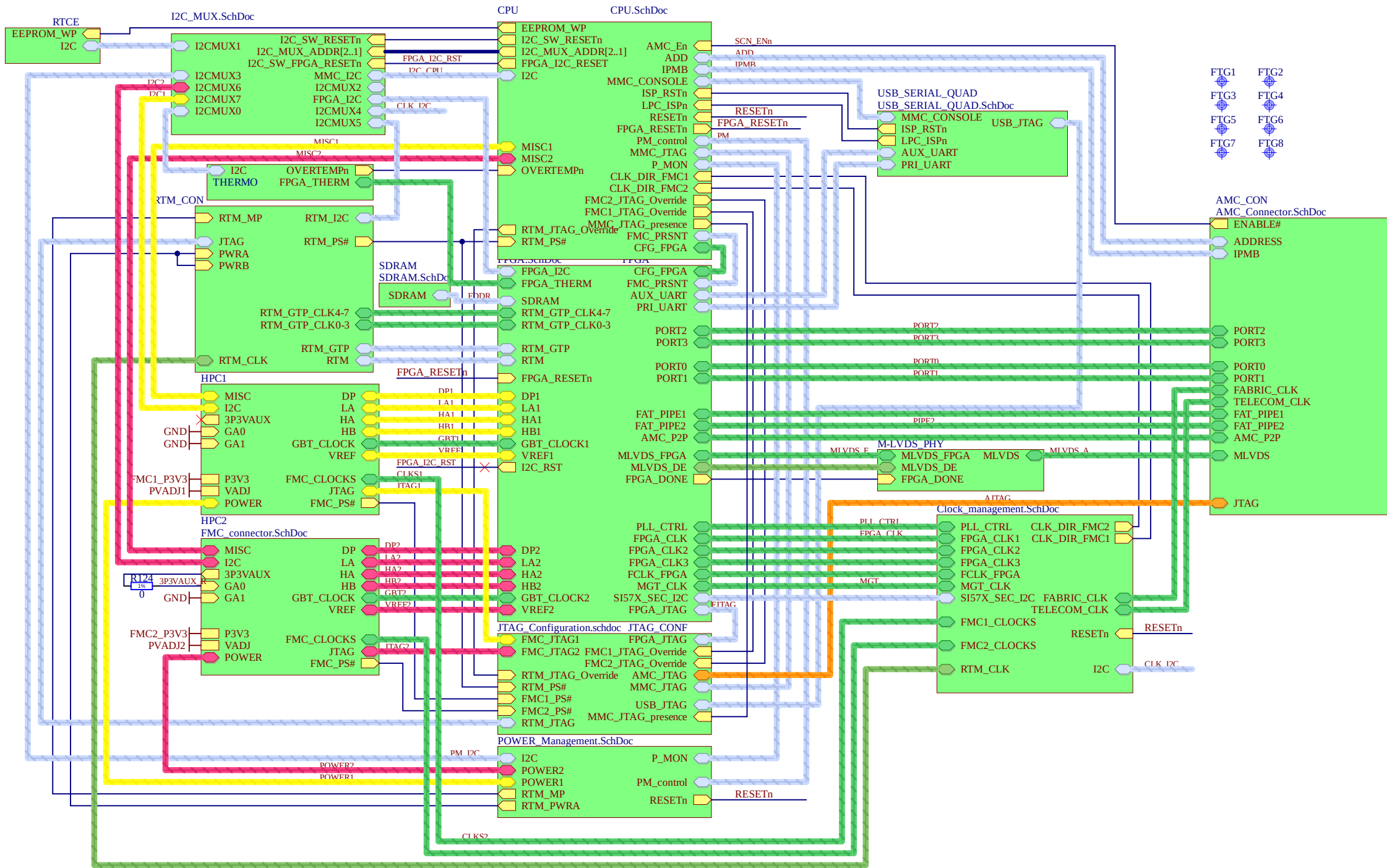
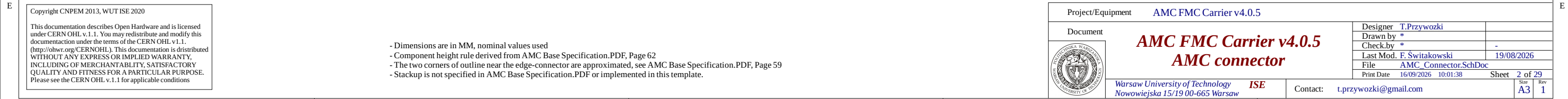


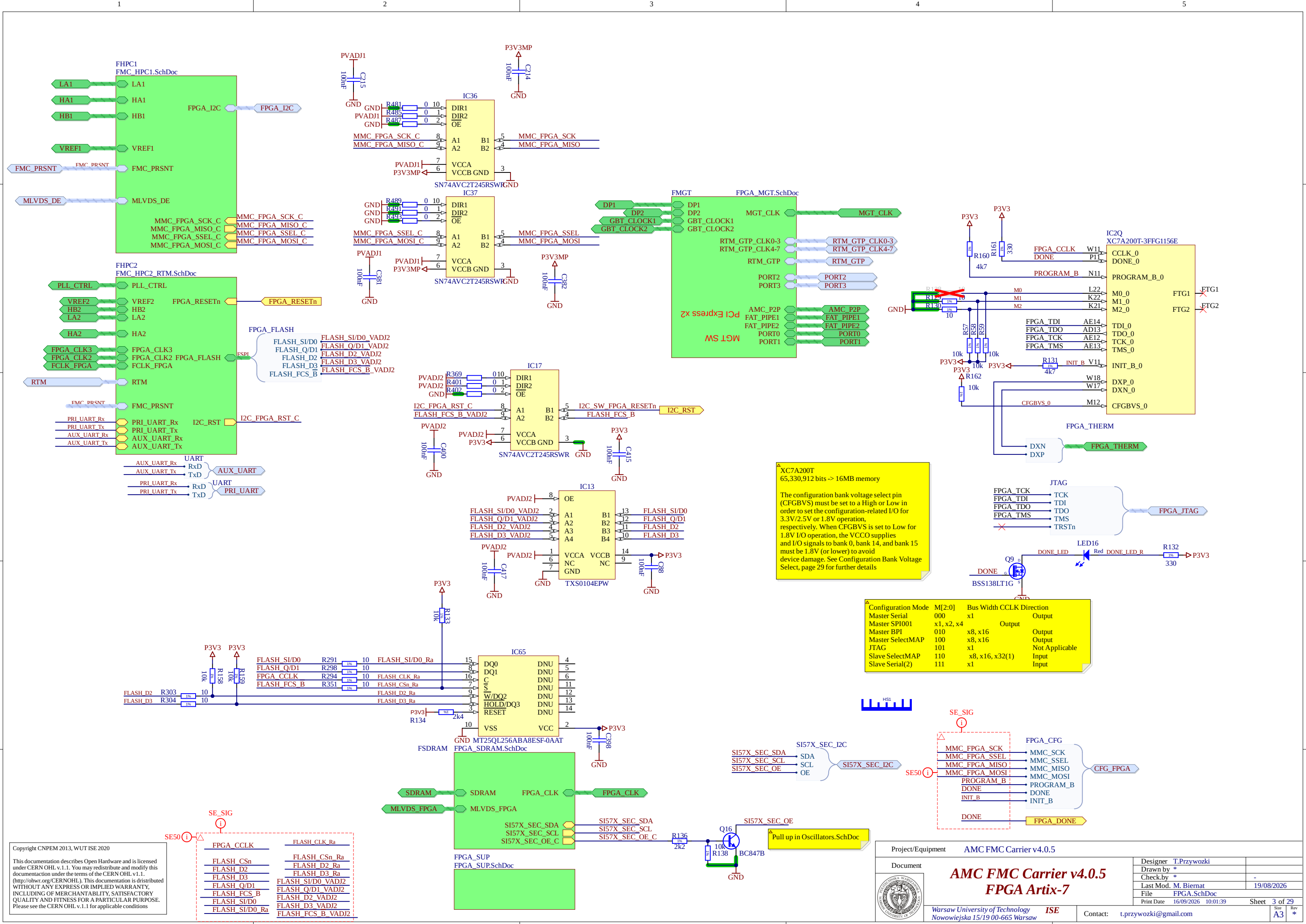


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Project/Equipment		AMC FMC Carrier v4.0.5	
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	Designer	T.Przywozki	
	Drawn by	*	XX/XX/XXXX
	Check by	*	-
	Last Mod.	M. Biernat	19/08/2026
File		AMC_FMC_Carrier.SchDoc	
Print Date		16/09/2026 10:01:38	Sheet 1 of 29
Contact:		t.przywozki@gmail.com	Size A3 Rev 1

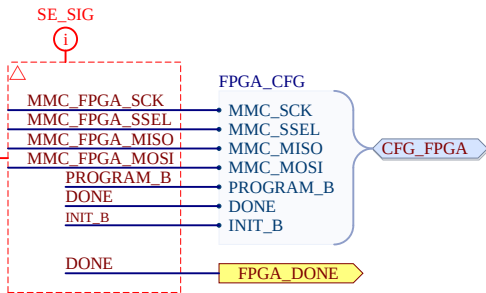




XC7A200T
65,330,912 bits -> 16MB memory

The configuration bank voltage select pin (CFGBVS) must be set to a High or Low in order to set the configuration-related I/O for 3.3V/2.5V or 1.8V operation, respectively. When CFGBVS is set to Low for 1.8V I/O operation, the VCCO supplies and I/O signals to bank 0, bank 14, and bank 15 must be 1.8V (or lower) to avoid device damage. See Configuration Bank Voltage Select, page 29 for further details

Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master Serial	000	x1	Output
Master SPI001	x1, x2, x4		Output
Master BPI	010	x8, x16	Output
Master SelectMAP	100	x8, x16	Output
JTAG	101	x1	Not Applicable
Slave SelectMAP	110	x8, x16, x32(1)	Input
Slave Serial(2)	111	x1	Input

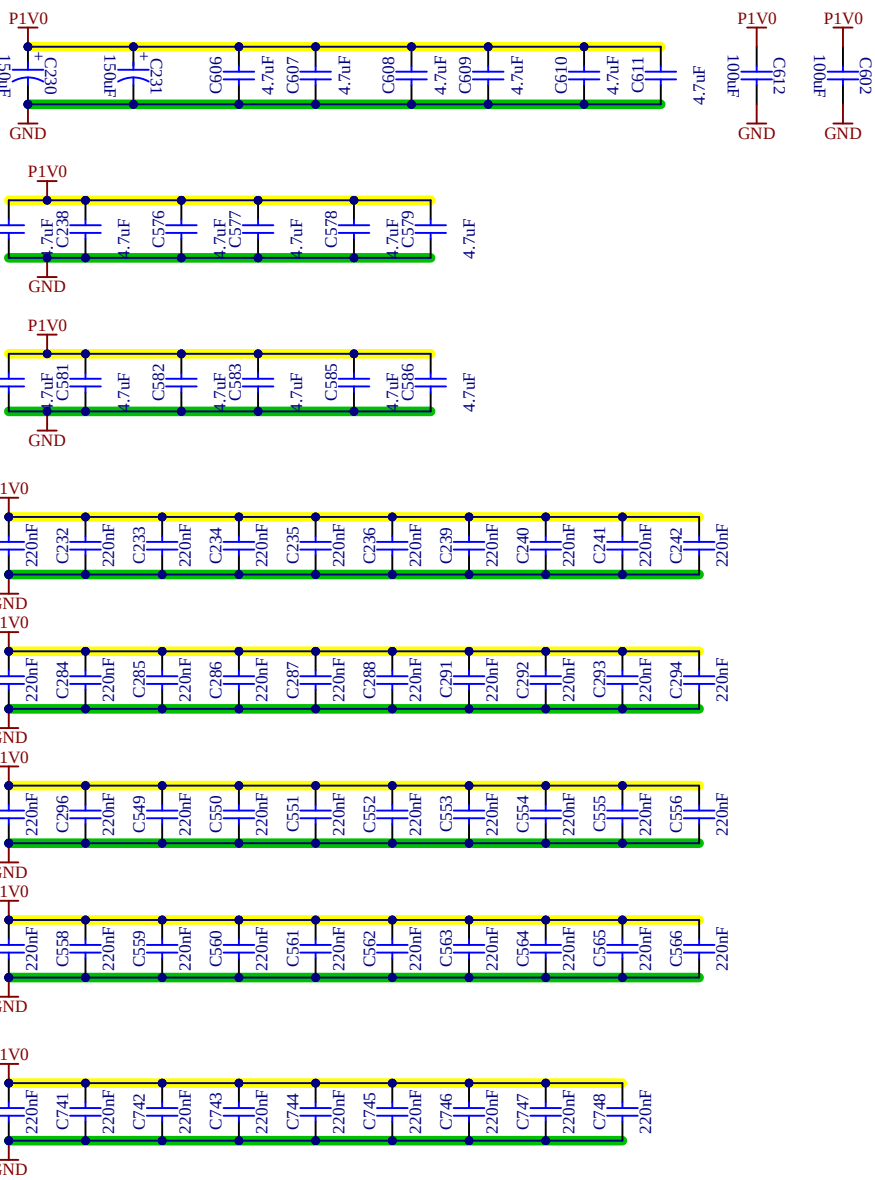


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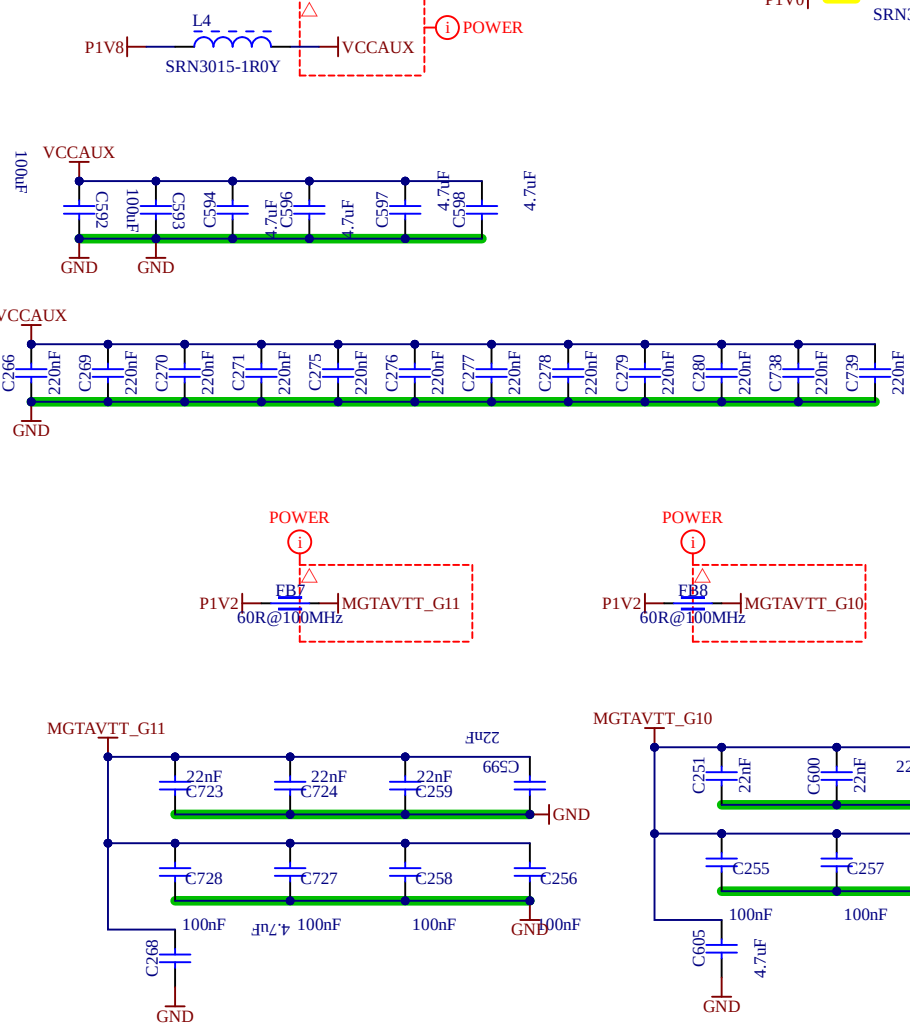
VCCINT

1x680uF, 2x100uF, 28x4.7uF
42x0.47uF on PG.34



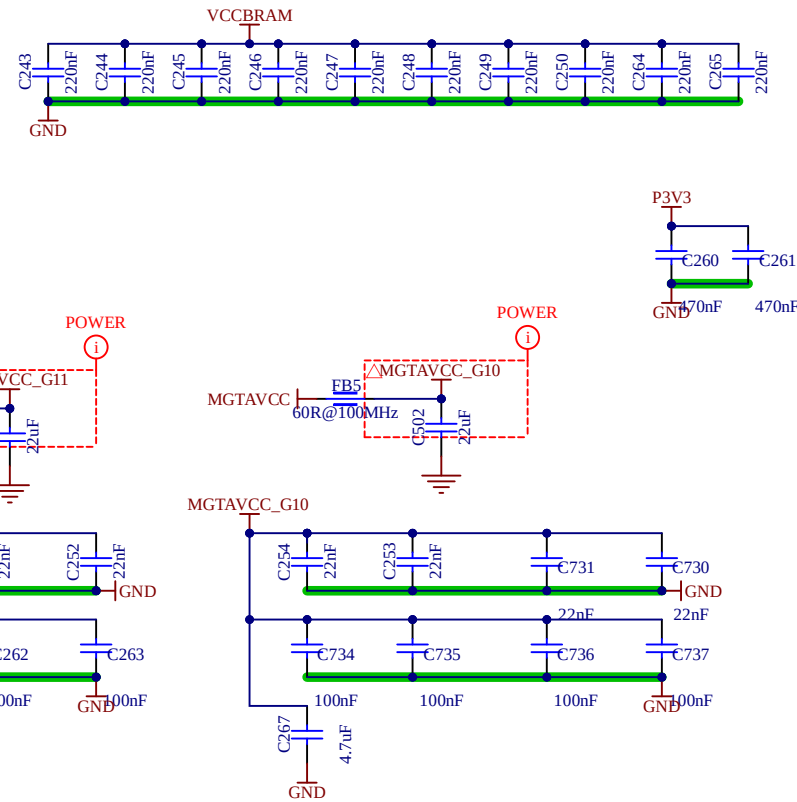
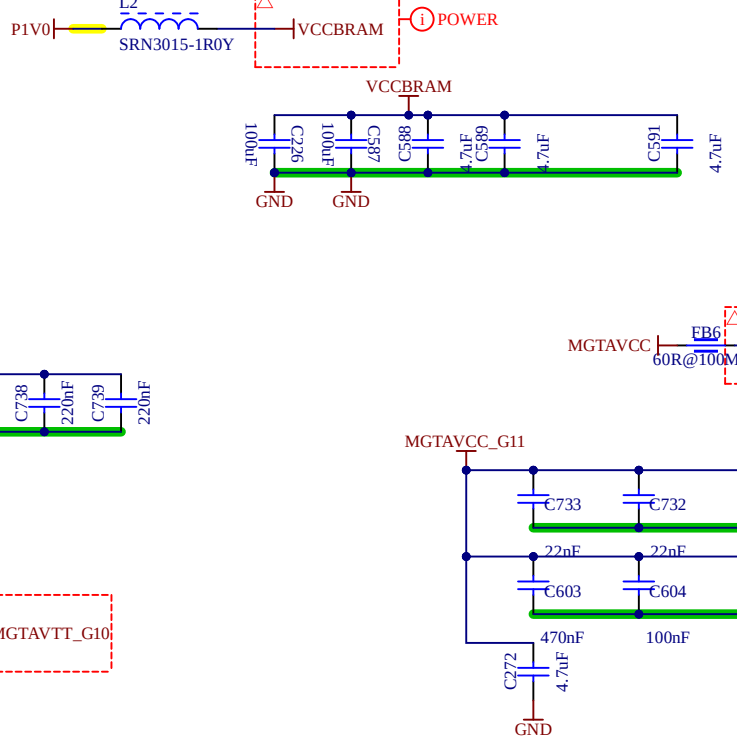
VCCAUX

3x47uF, 4x4.7uF, 7x0.47uF



VCCBRAM

2x100uF, 4x4.7uF, 8x0.47uF



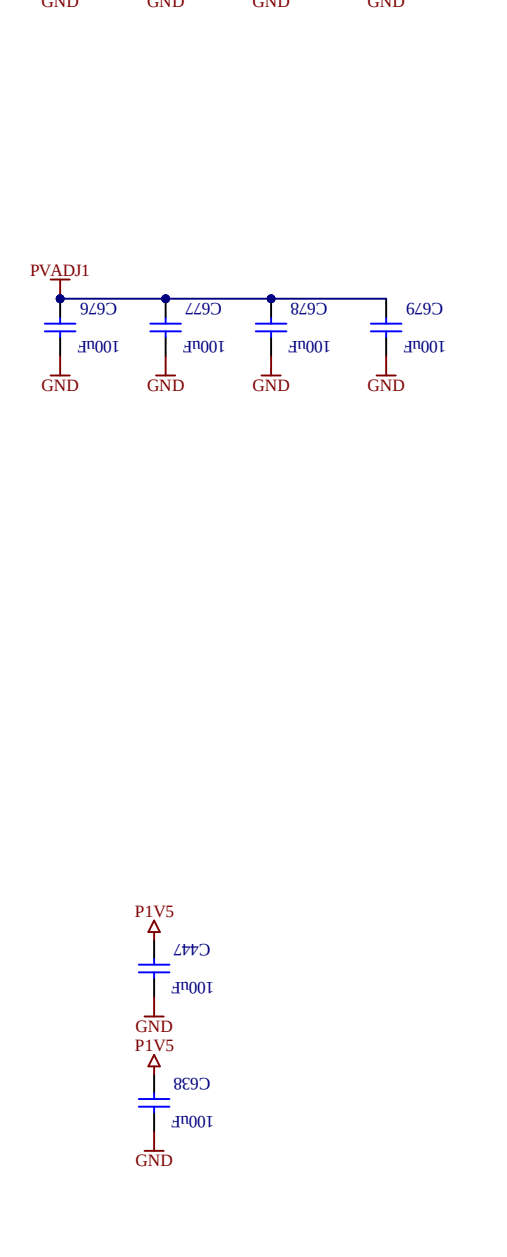
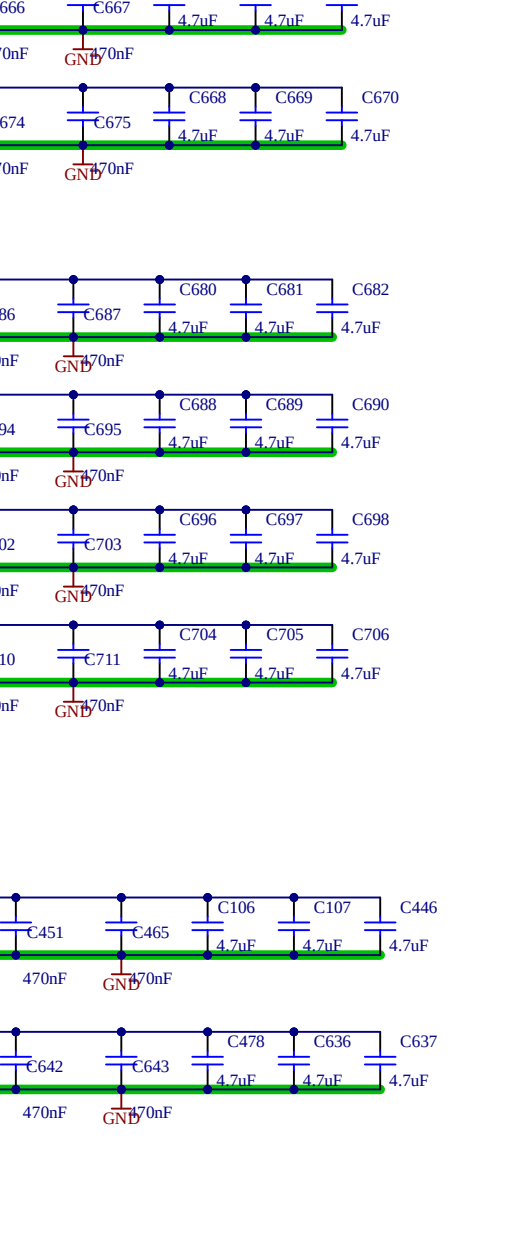
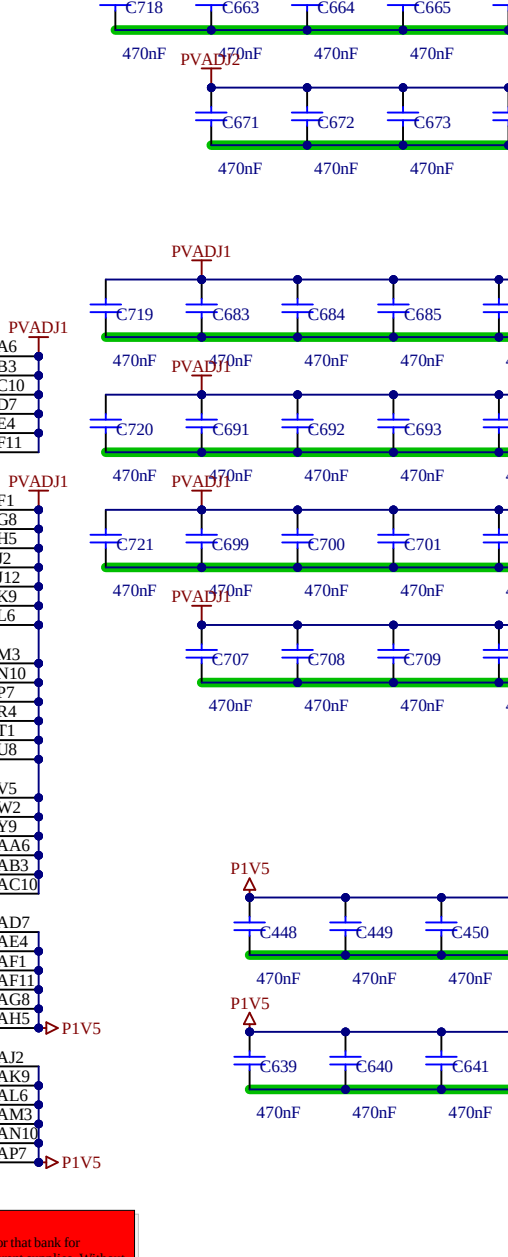
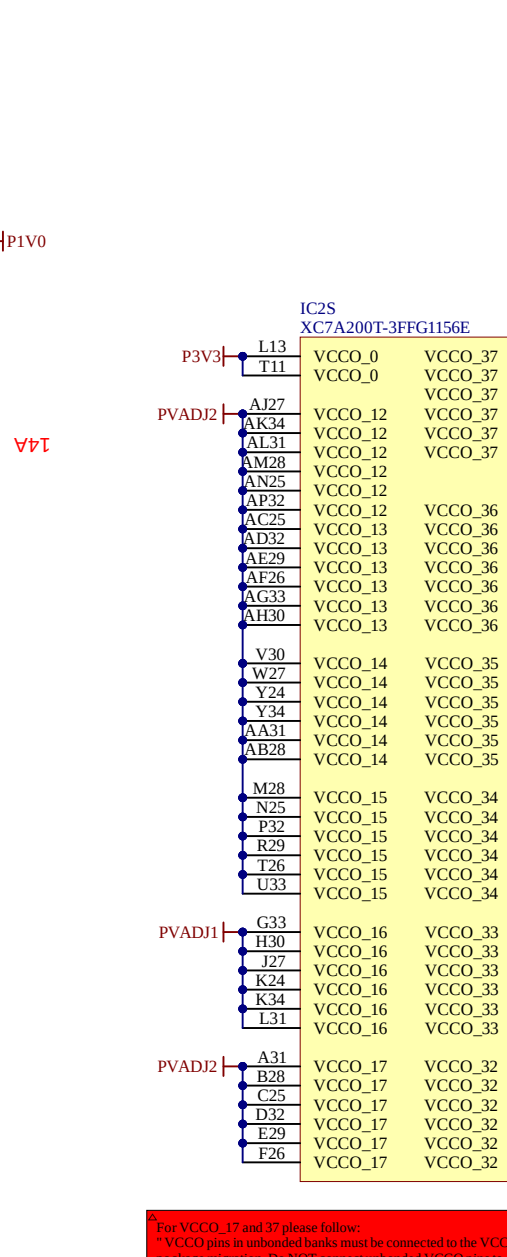
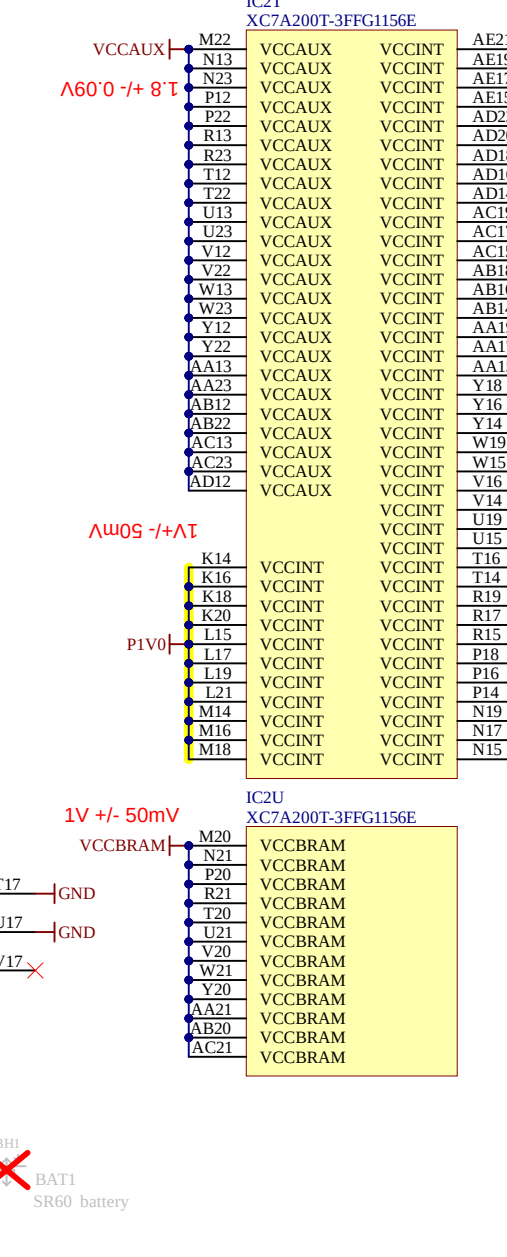
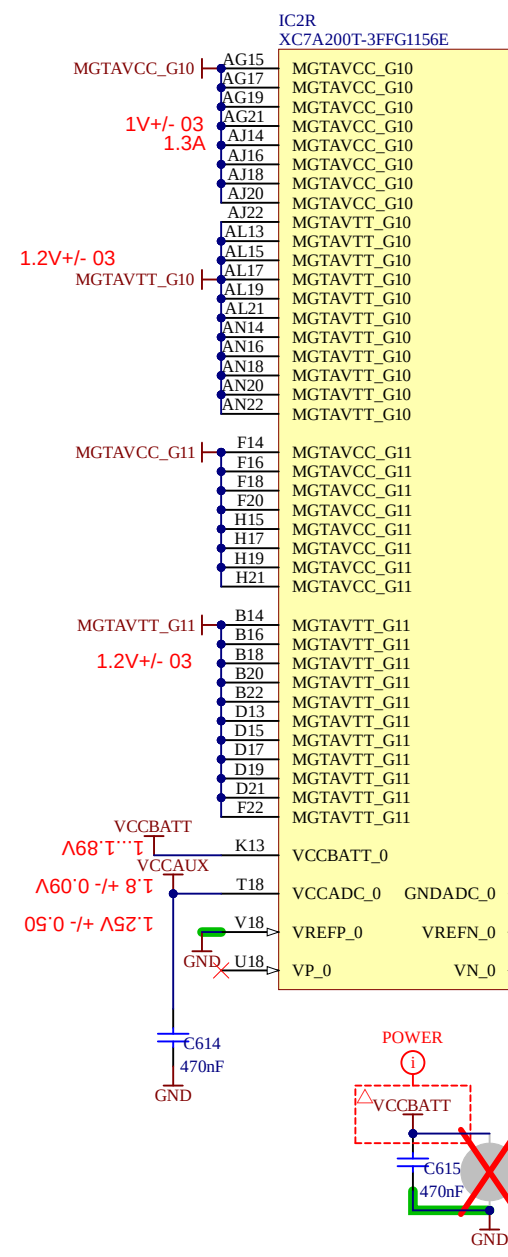
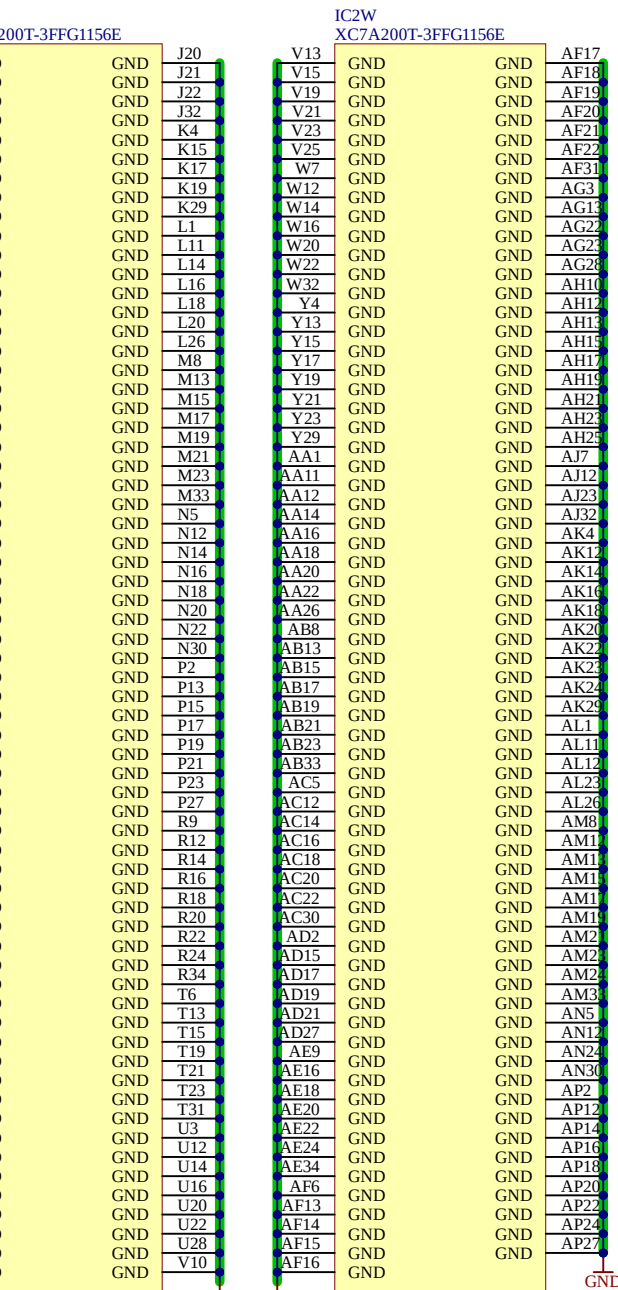
MGTAVCC	1V	2.3A	
MGTAVTT	1.2V	1.5A	
MGTVCCAUX	1.8V	0.5A	
VCCAUX	1.8V	1A	seq1
VCCAUX_IO	1.8/2V	0.1A	seq1
VCCBRAM	1V	1.8A	seq0
VCCINT	1V	6A	seq0
VCCO_1,2	1.2V	3.2A	seq2
VCCO_1,35	1.35V	0.9A	seq2
VCCO_1,5	1.5V	0.9A	seq2
VCCO_1,8	1.8V	0.9A	seq2
VCCO_2,5	2.5V	0.9A	seq2
VCCO_3,3	3.3V	0.9A	seq2

power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX

power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT OR VMGTAVCC, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTAVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.

When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT - VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.

When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.



For VCCO_17 and 12 please follow:
VCCO pins in unbonded banks must be connected to the VCCO for that bank for package migration. Do NOT connect unbonded VCCO pins to different supplies. Without package migration requirement, VCCO pins in unbonded banks can be tied to a common supply (VCCO or ground).
No migration available (since there is no 350T) - I recommend connecting to empty power rail and using as power decoupling placeholder for decaps. Remove bulk C22 and C29

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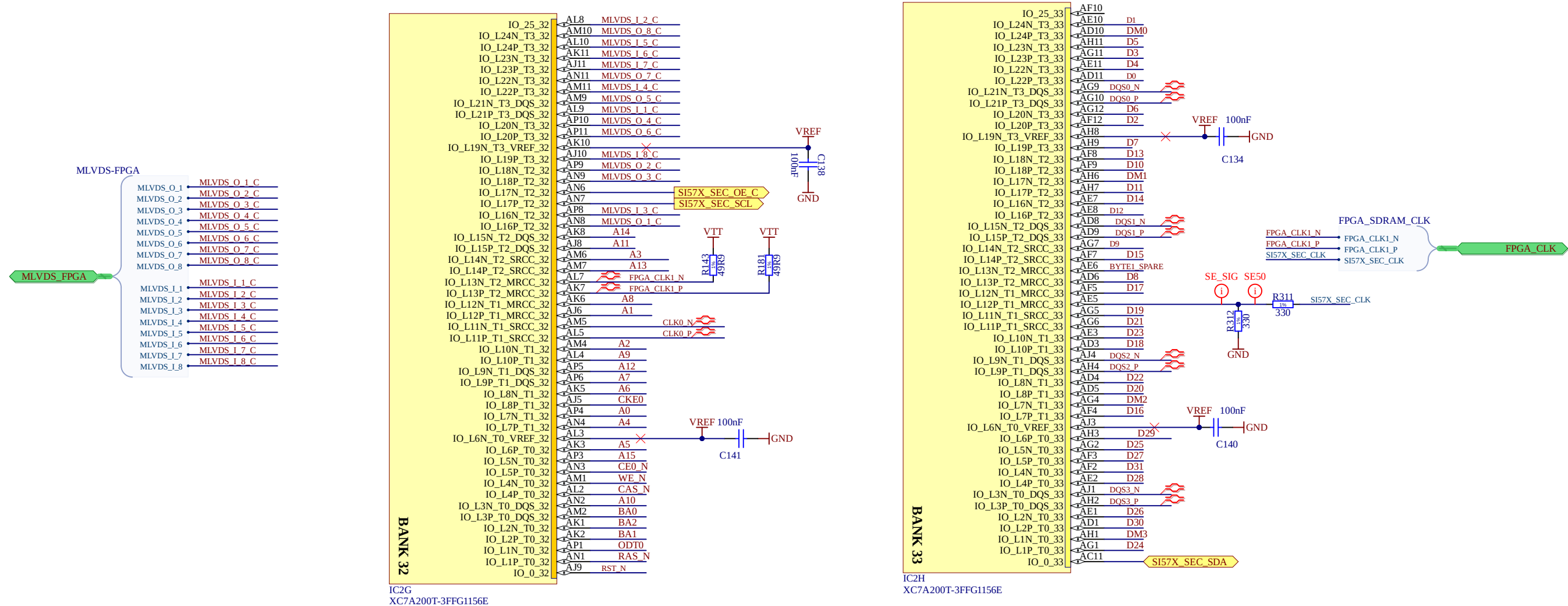
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		Check by	*
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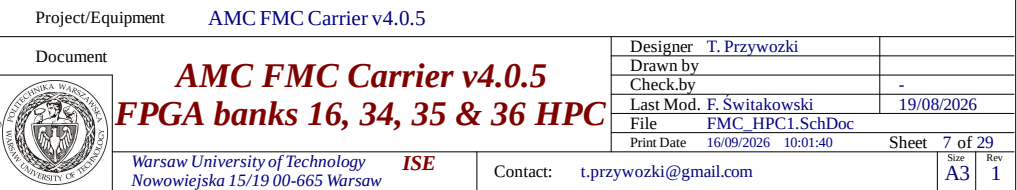


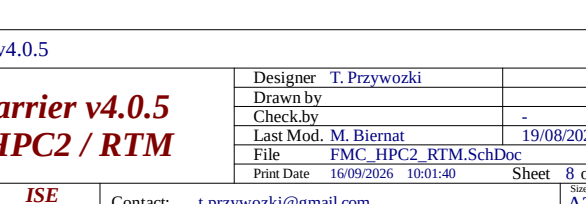
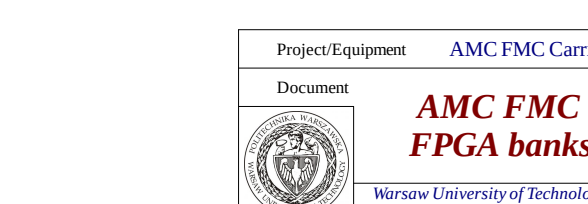
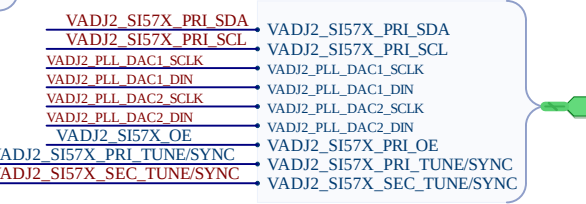
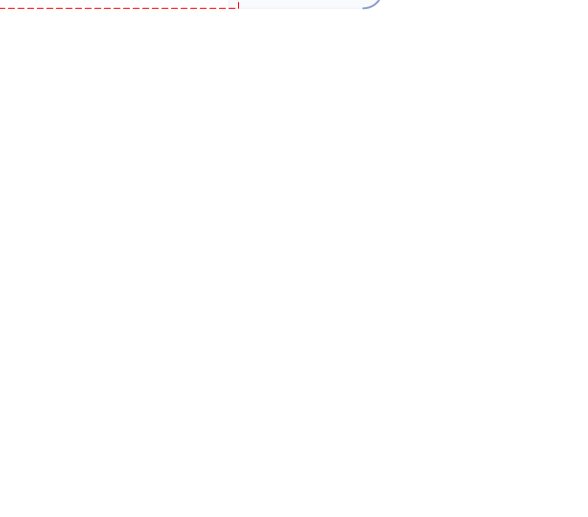
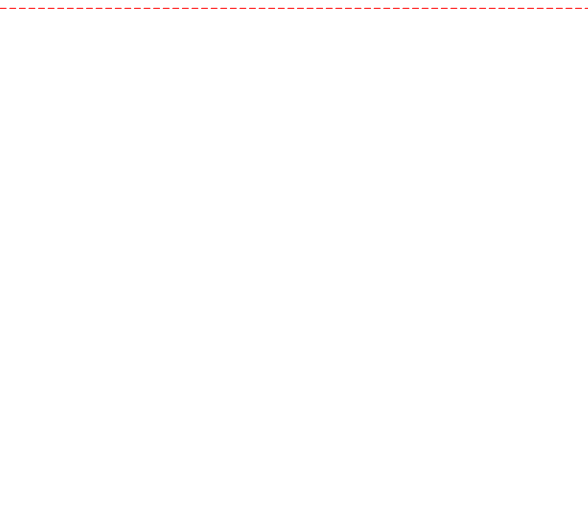
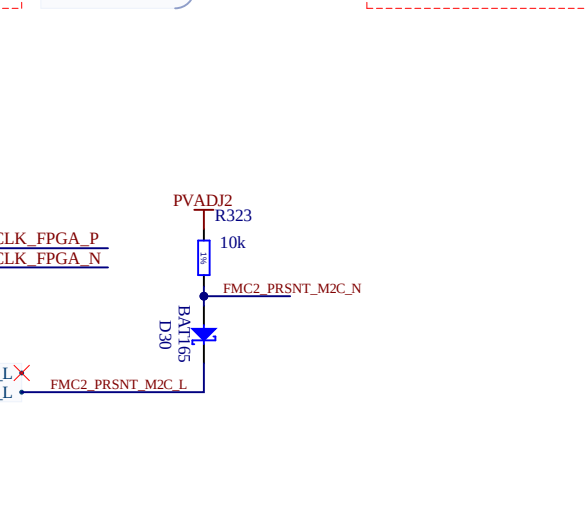
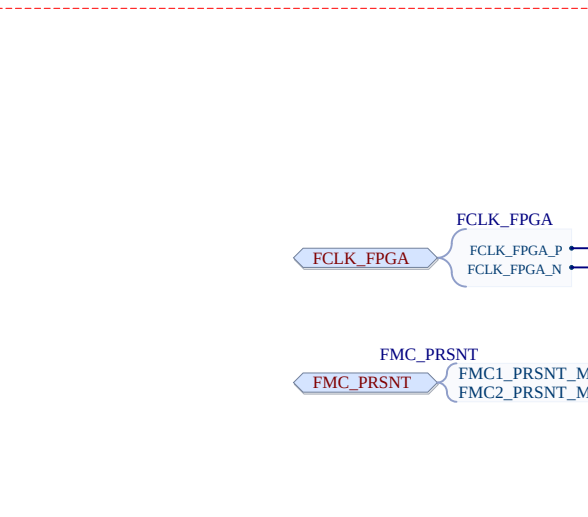
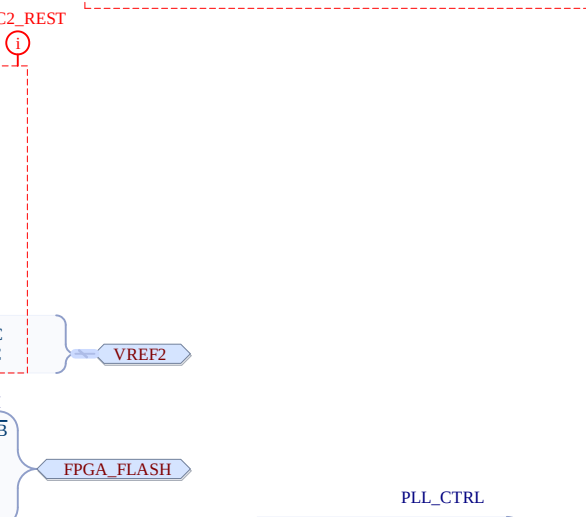
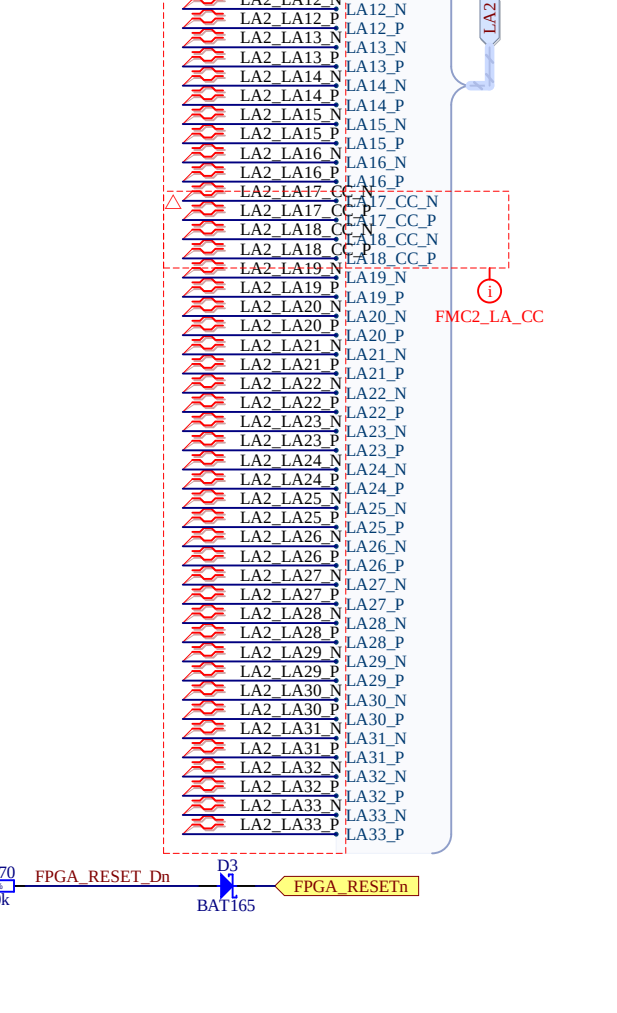
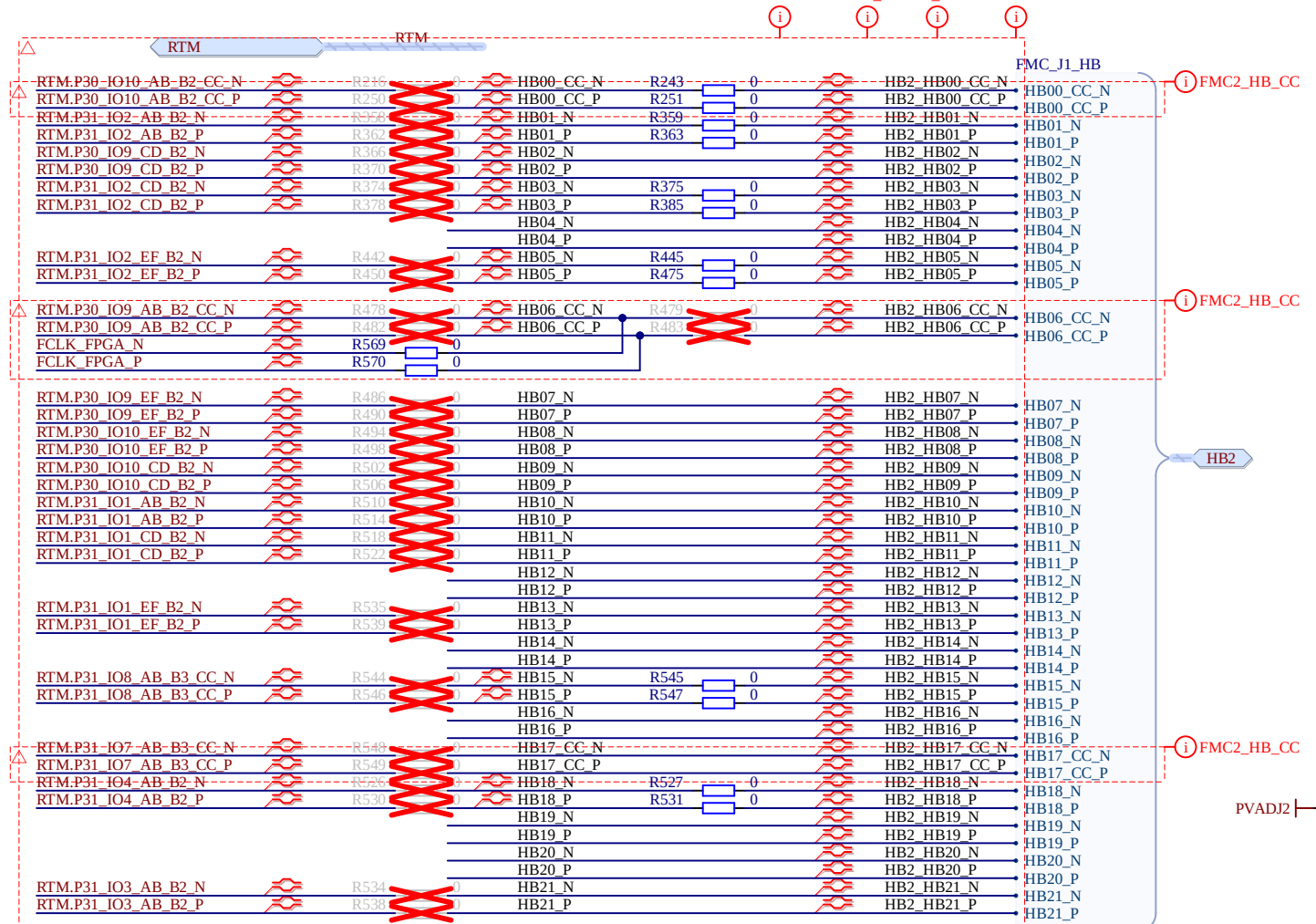
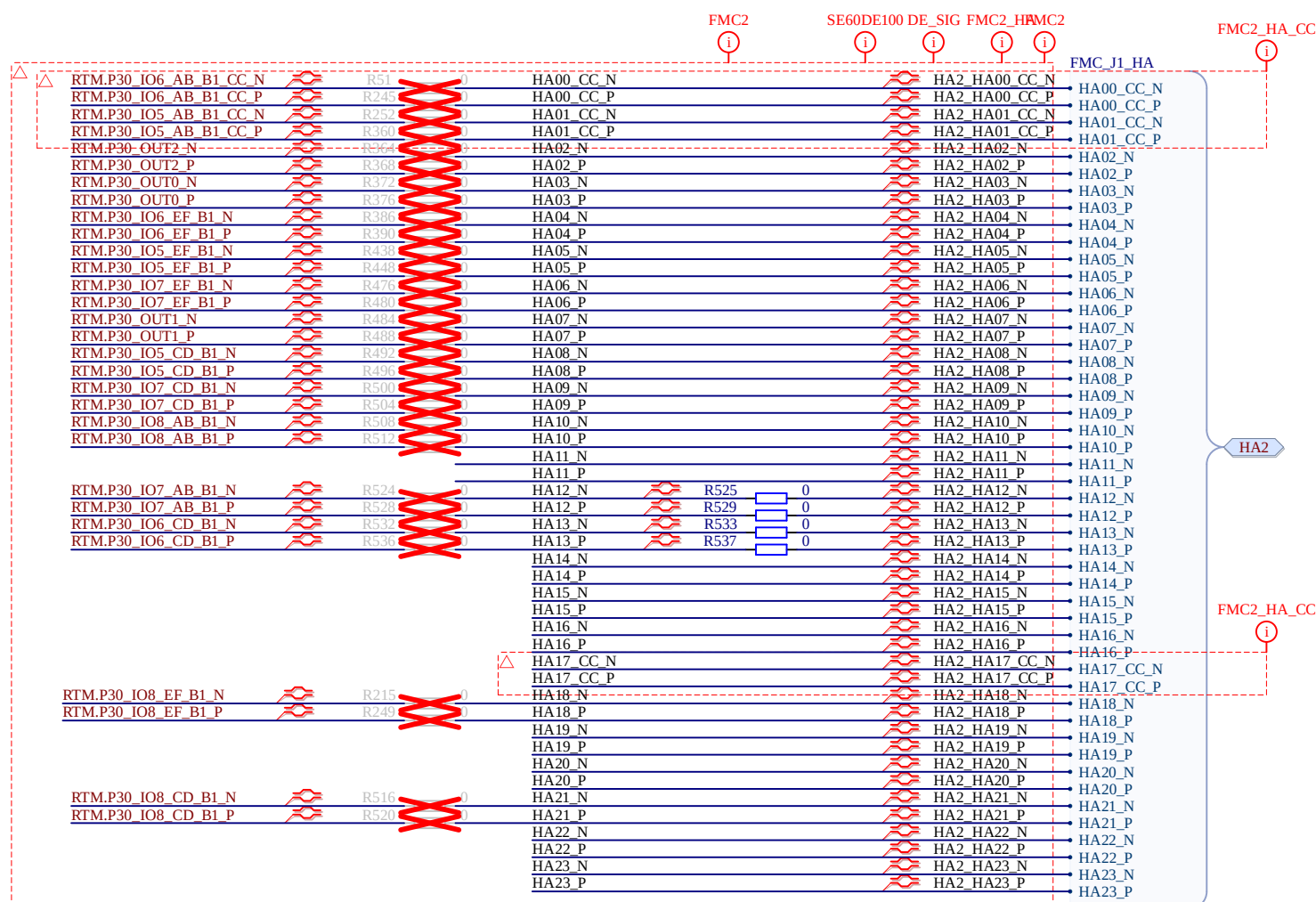
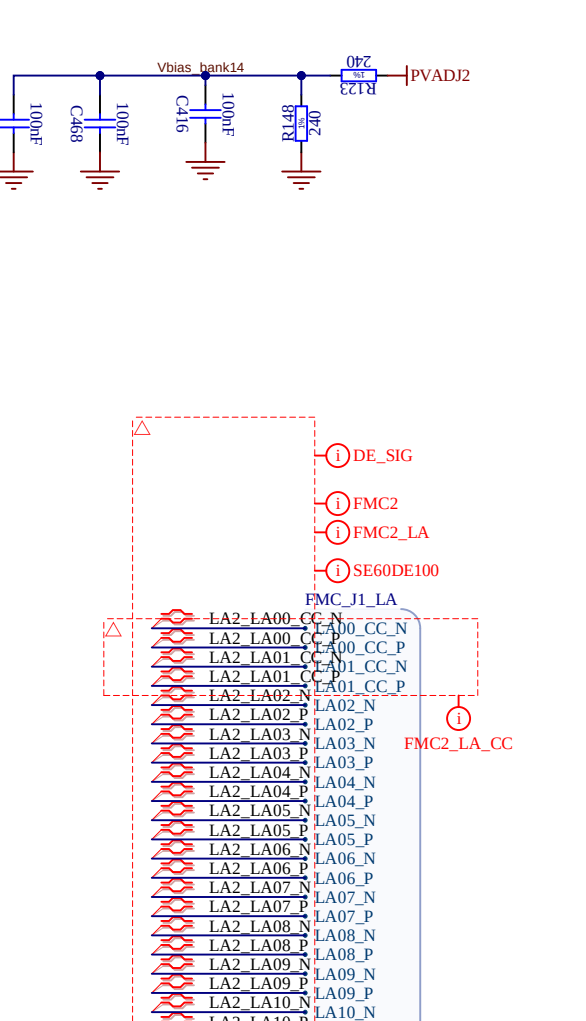
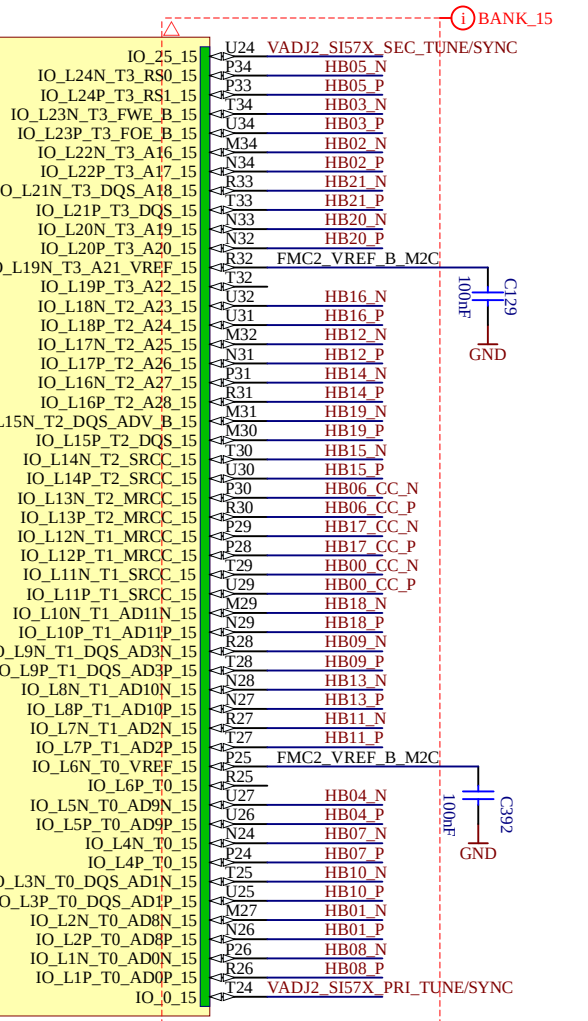
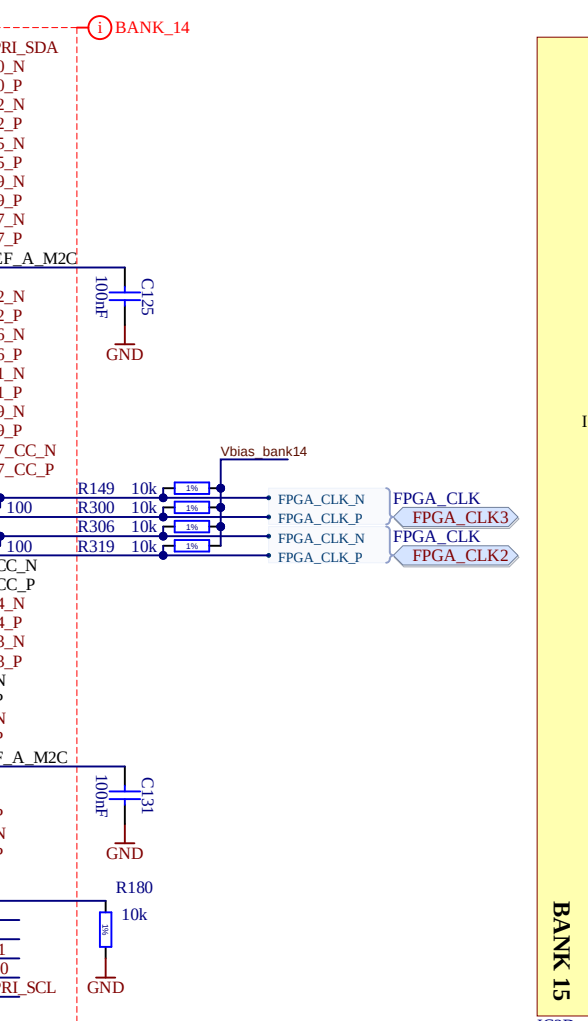
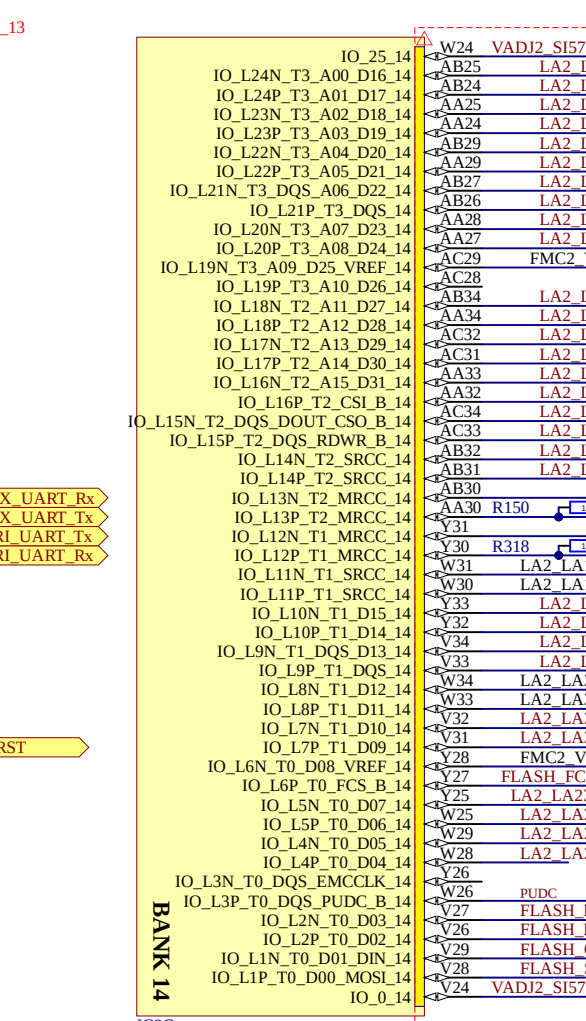
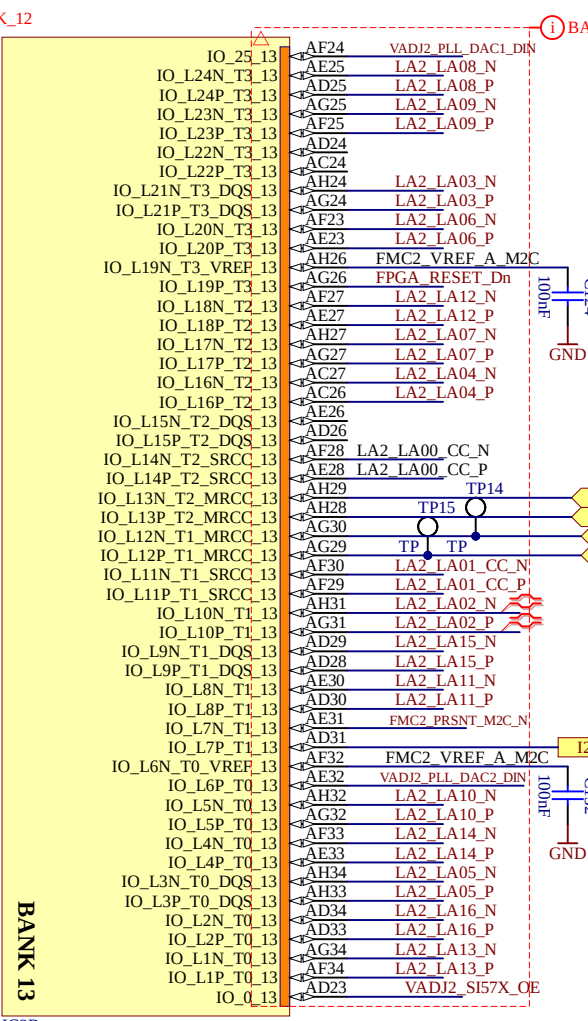
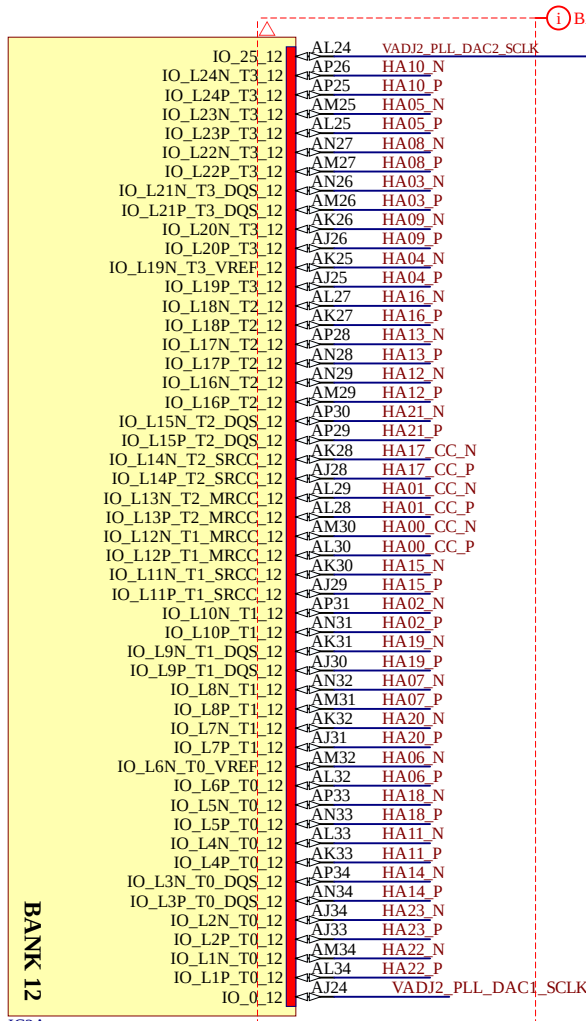
AMC FMC Carrier v4.0.5
FPGA Power & Decoupling

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Nowowiejska 15/19 00-665 Warsaw

ISE
Contact: t.przywowski@gmail.com

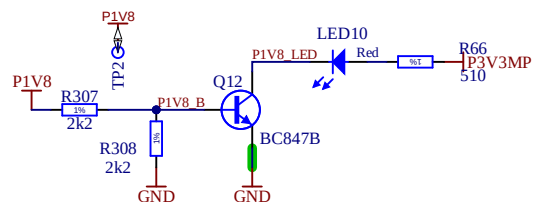
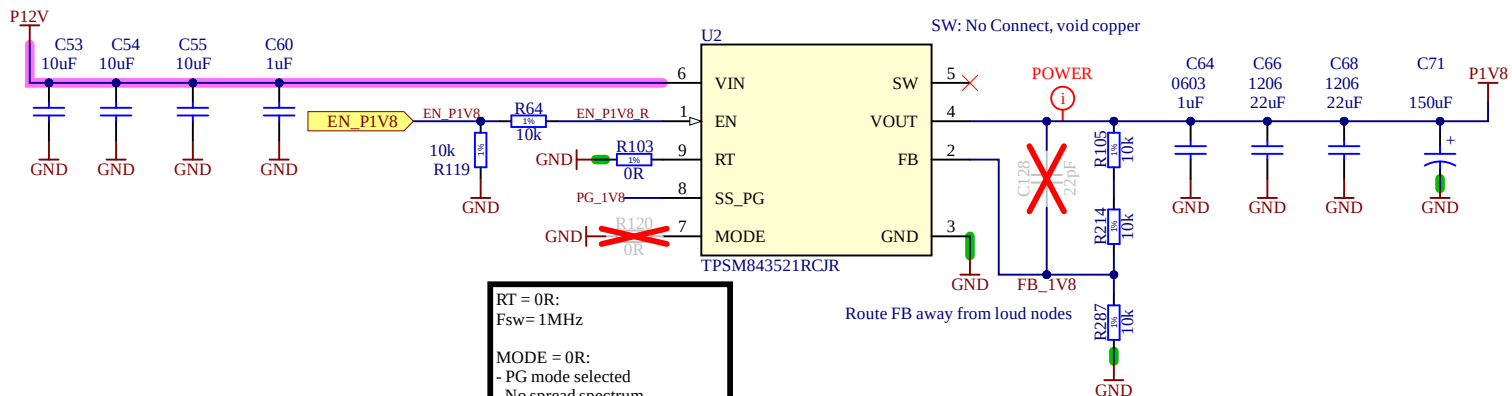
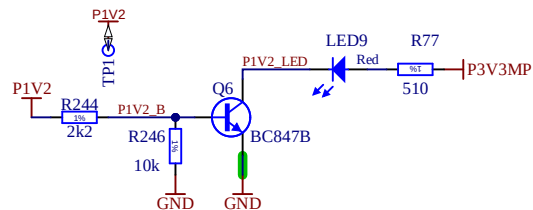
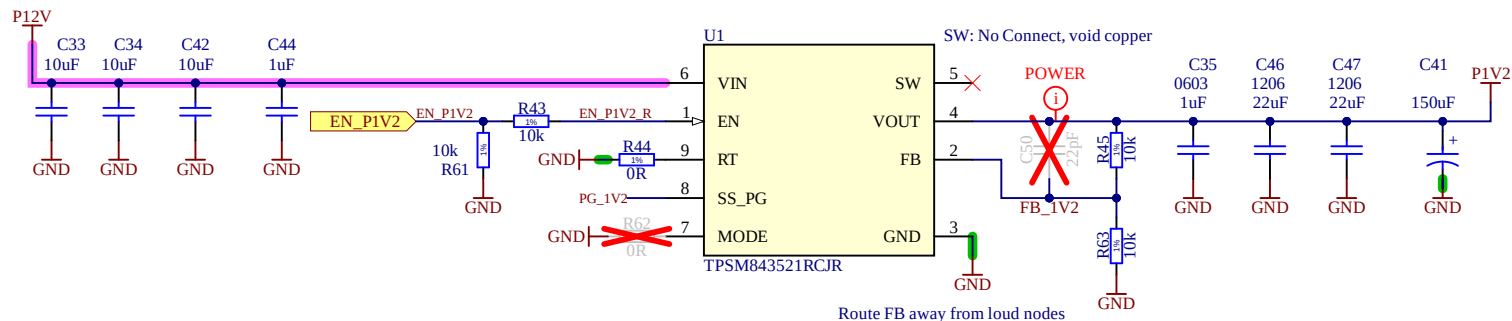








Compatible parts:
TPSM843521 - 5A
TPSM843321 - 3A
TPSM84538 - 5A, 28V in
TPSM84338 - 3A, 28V in

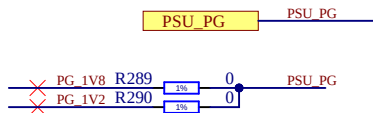


RT = 0R:
Fsw = 1MHz

MODE = 0R:
- PG mode selected
- No spread spectrum
- FCCS mode

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Project/Equipment AMC FMC Carrier v4.0.5

Document

AMC FMC Carrier v4.0.5
Power P1V2 & P1V8



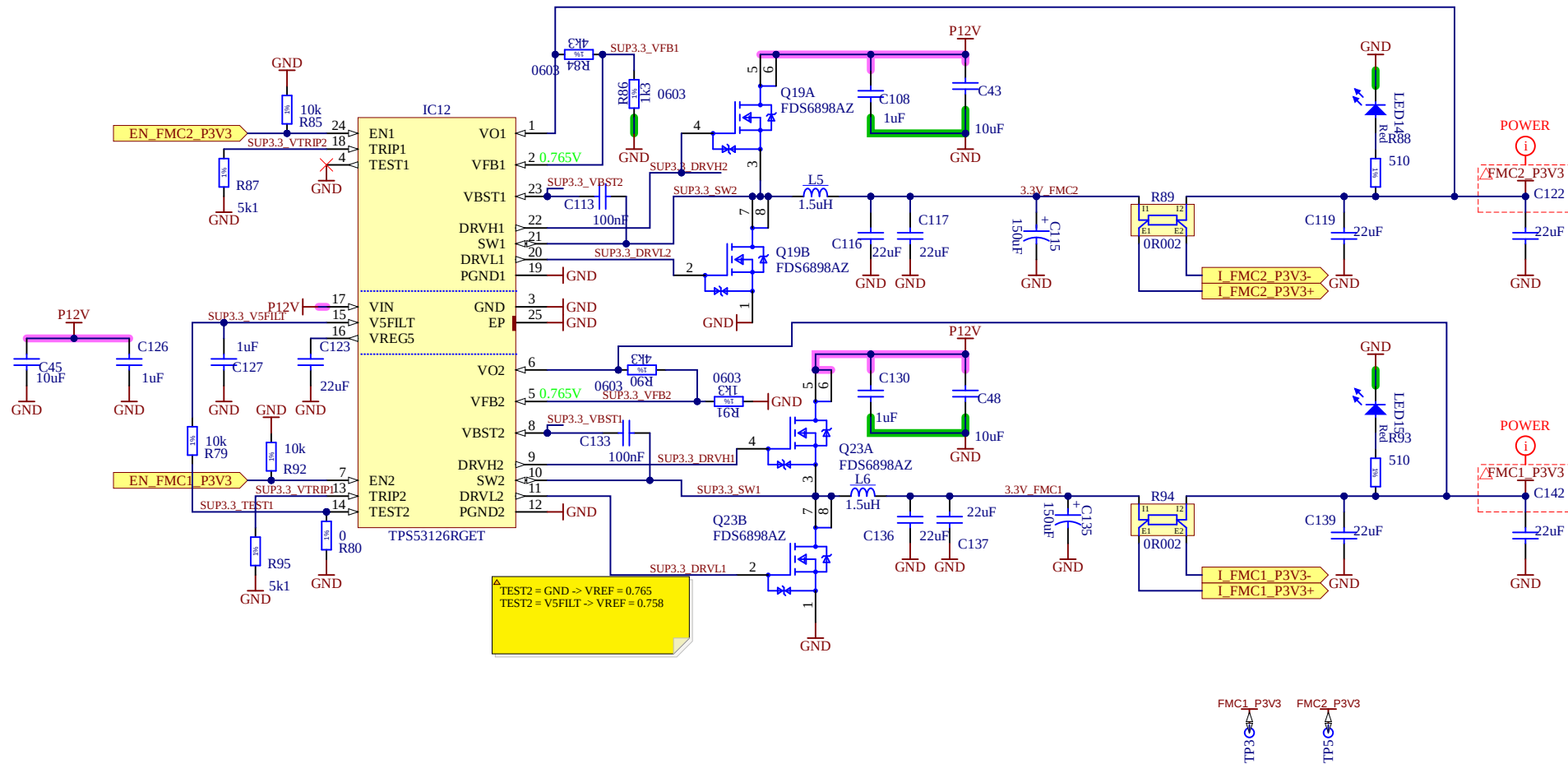
Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

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File	SUP_1.2_1.8.SchDoc	
Print Date	16/09/2026 10:01:41	Sheet 11 of 29

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Project/Equipment AMC FMC Carrier v4.0.5

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AMC FMC Carrier v4.0.5 Power FMCs P3V3



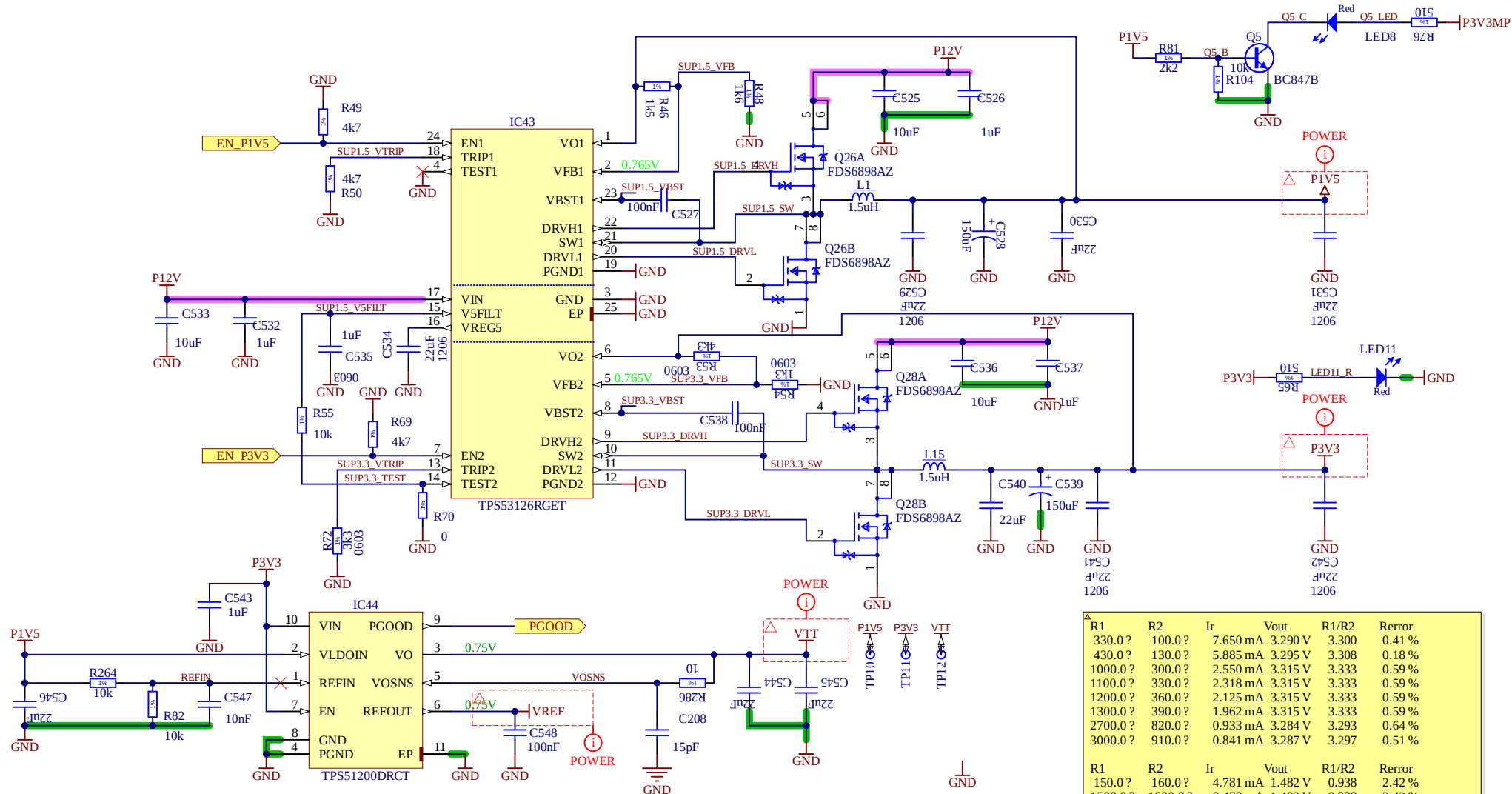
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Project/Equipment **AMC FMC Carrier v4.0.5**

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Size	Rev
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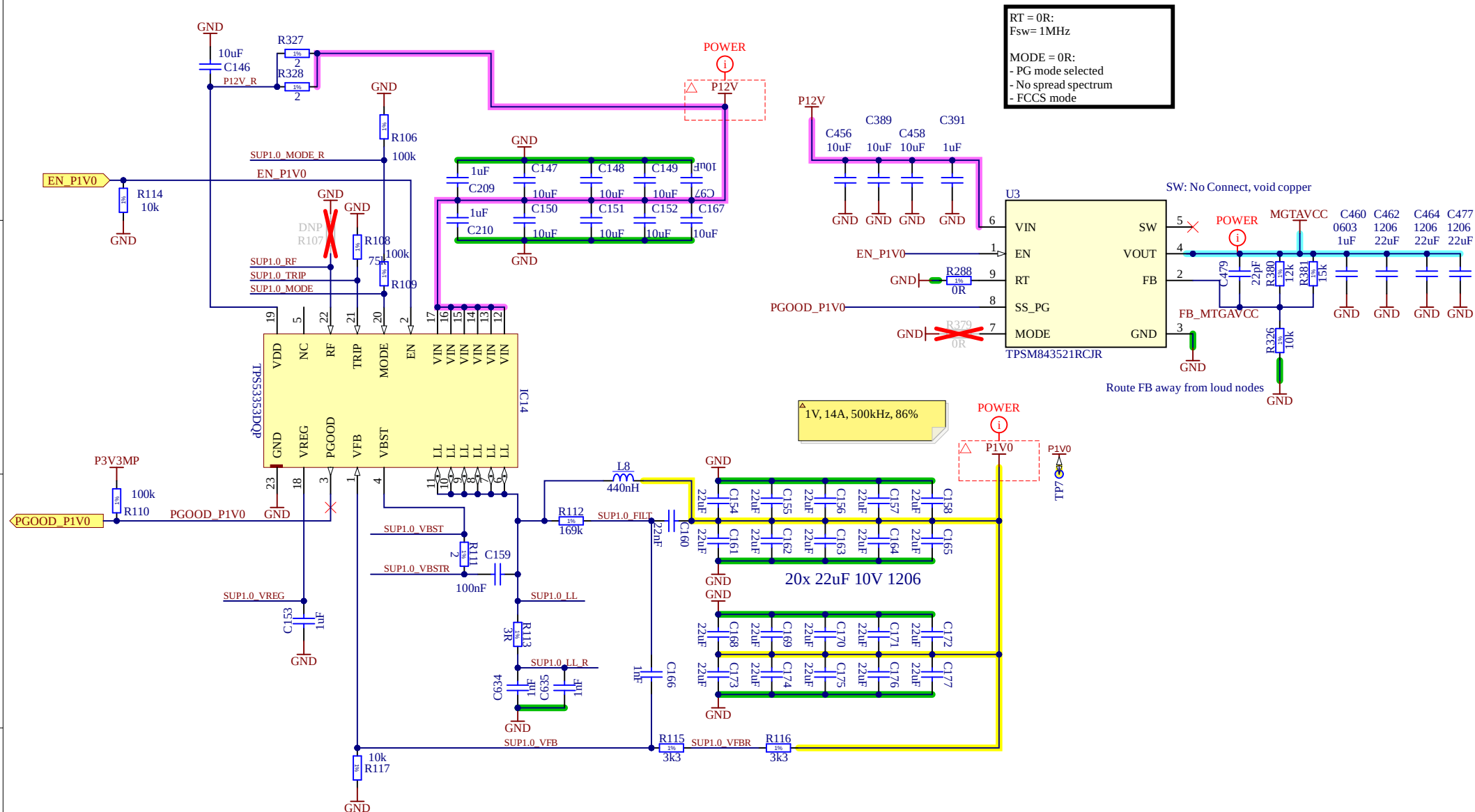
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A

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C

D



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Project/Equipment AMC FMC Carrier v4.0.5

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Power P1V0



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Size A4
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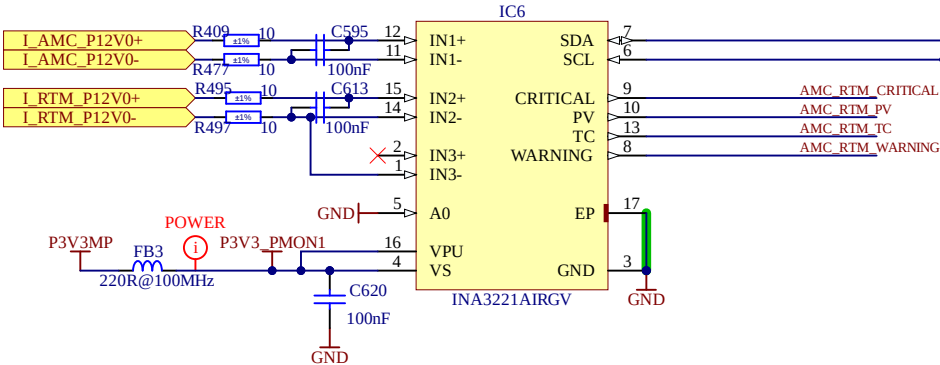
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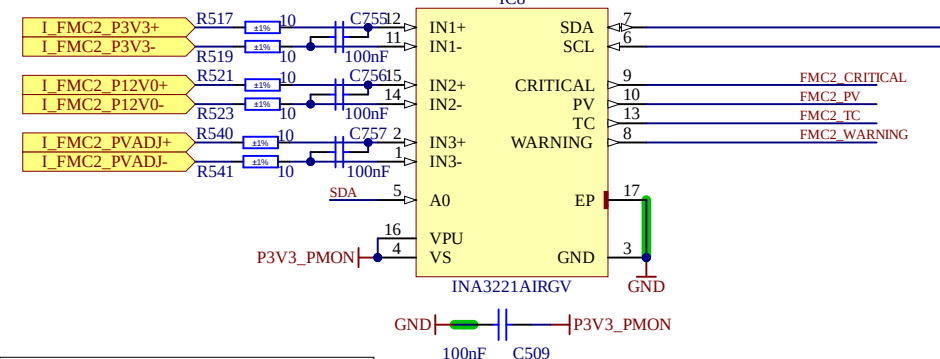
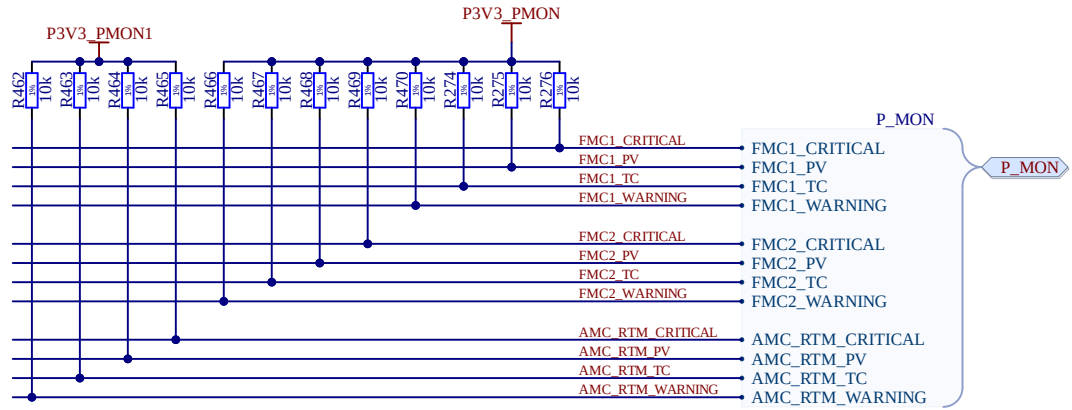
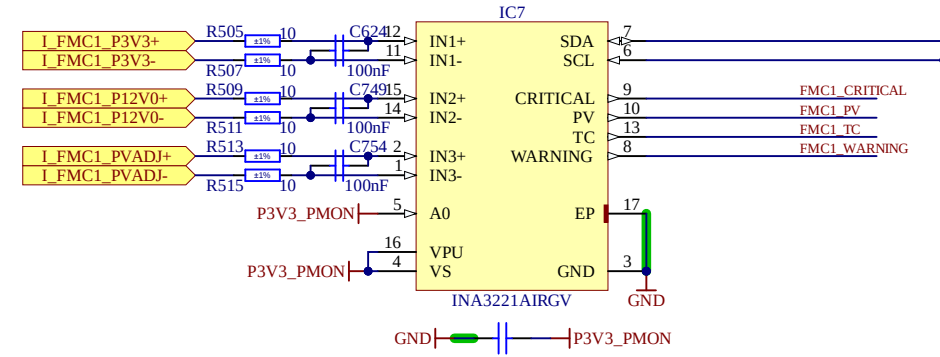
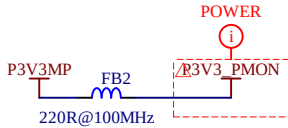
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C

D



A1	SLAVE ADDRESS
GND	1000000
VS	1000001
SDA	1000010
SCL	1000011



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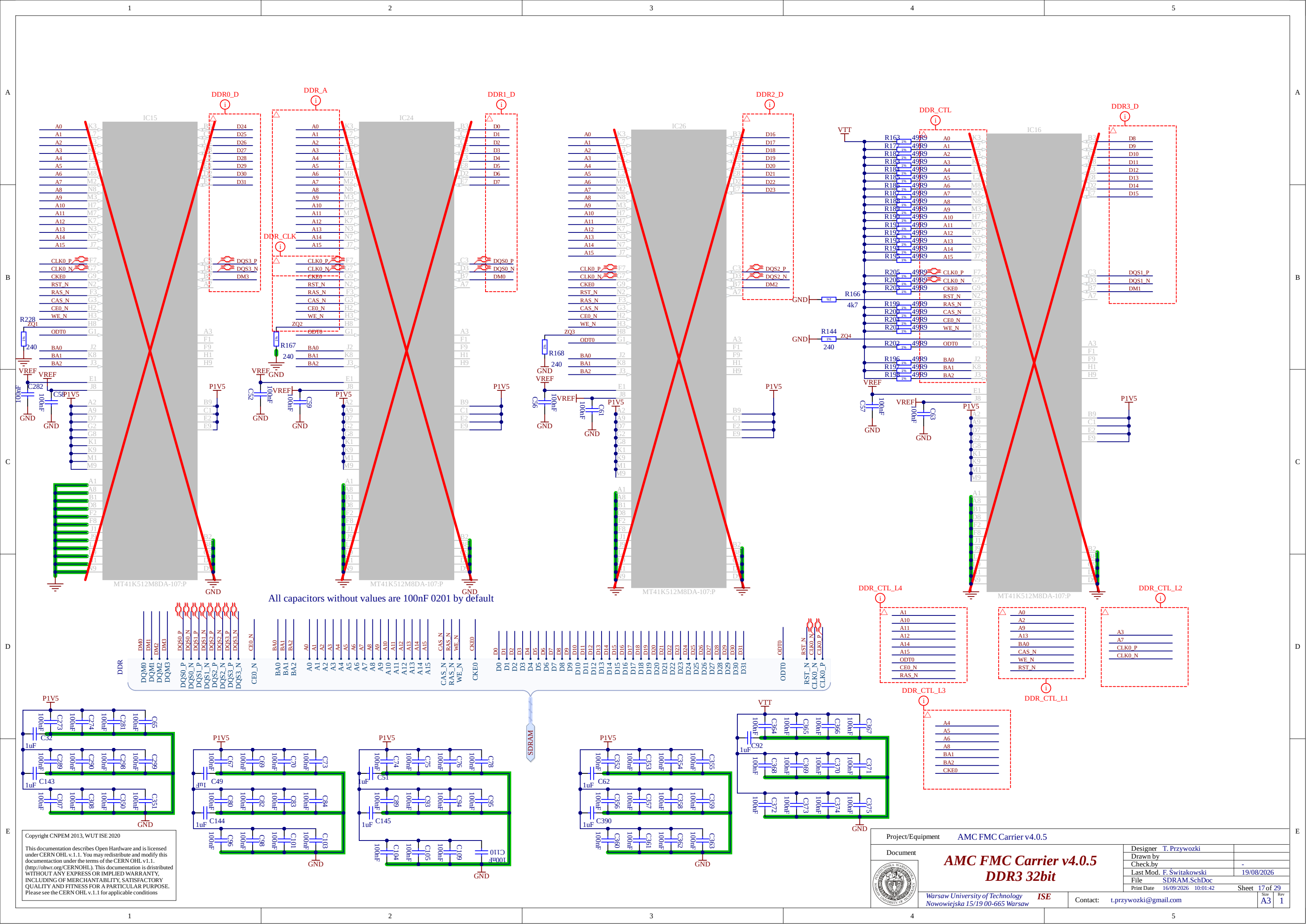
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Rev	1	Contact:	t.przywozki@gmail.com

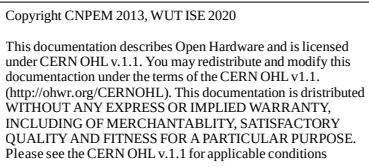


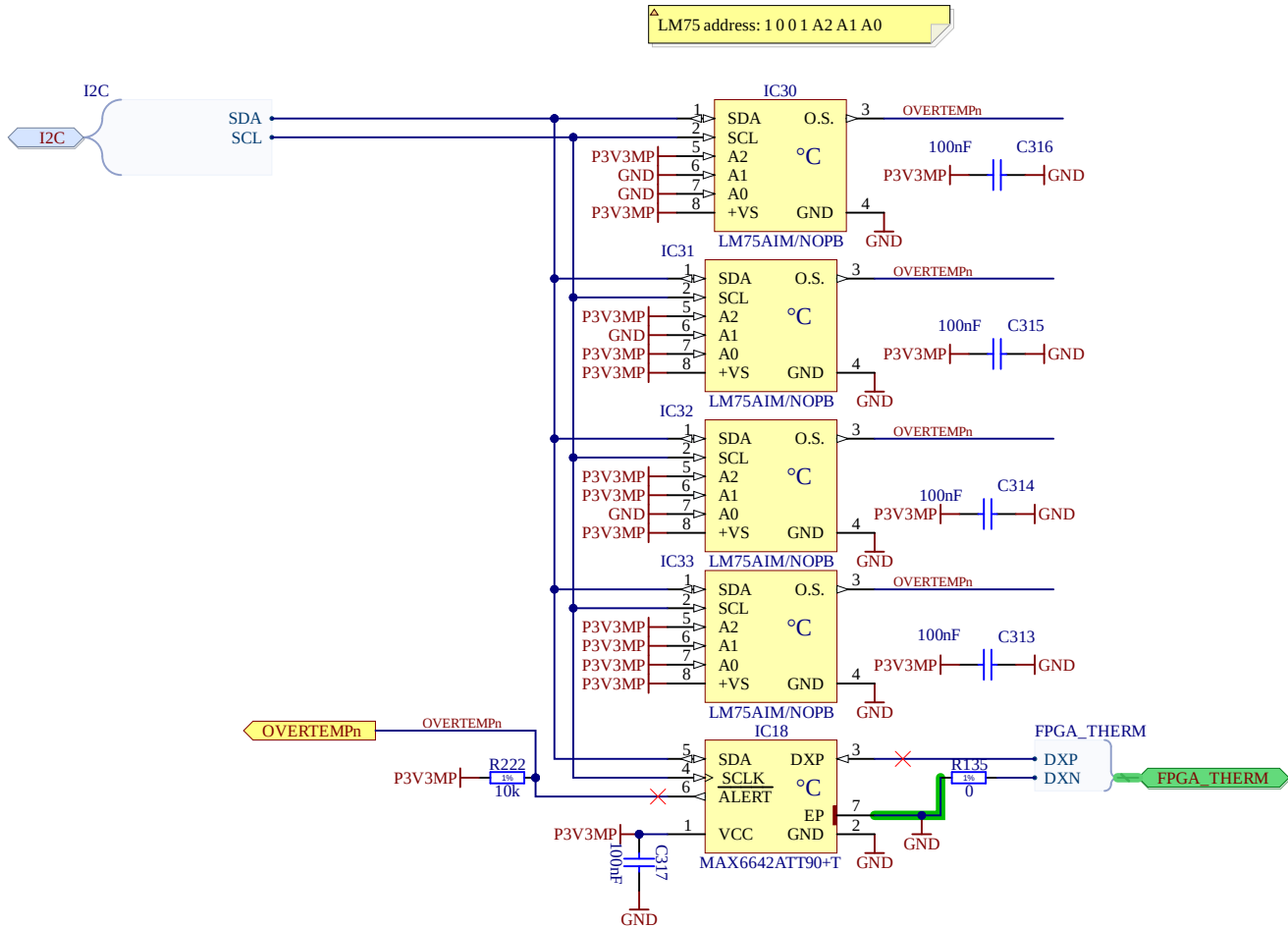
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Nowowiejska 15/19 00-665 Warsaw

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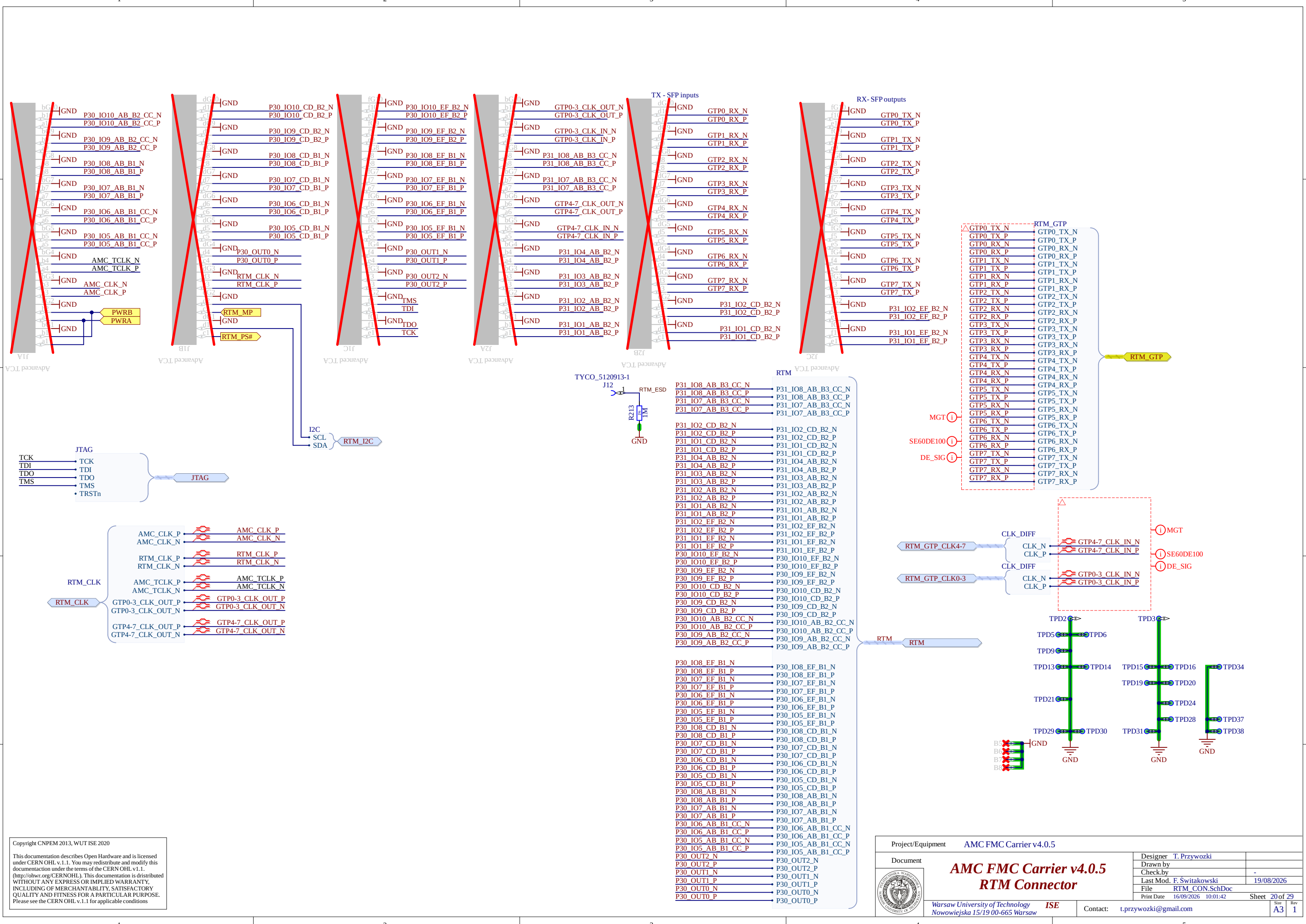




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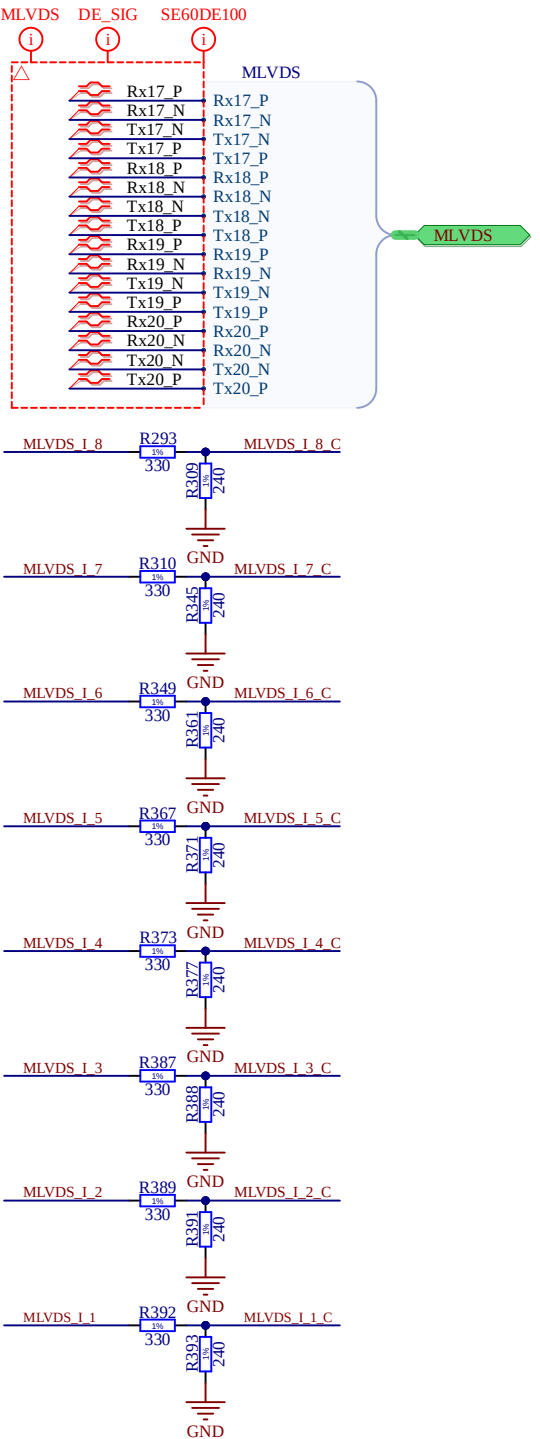
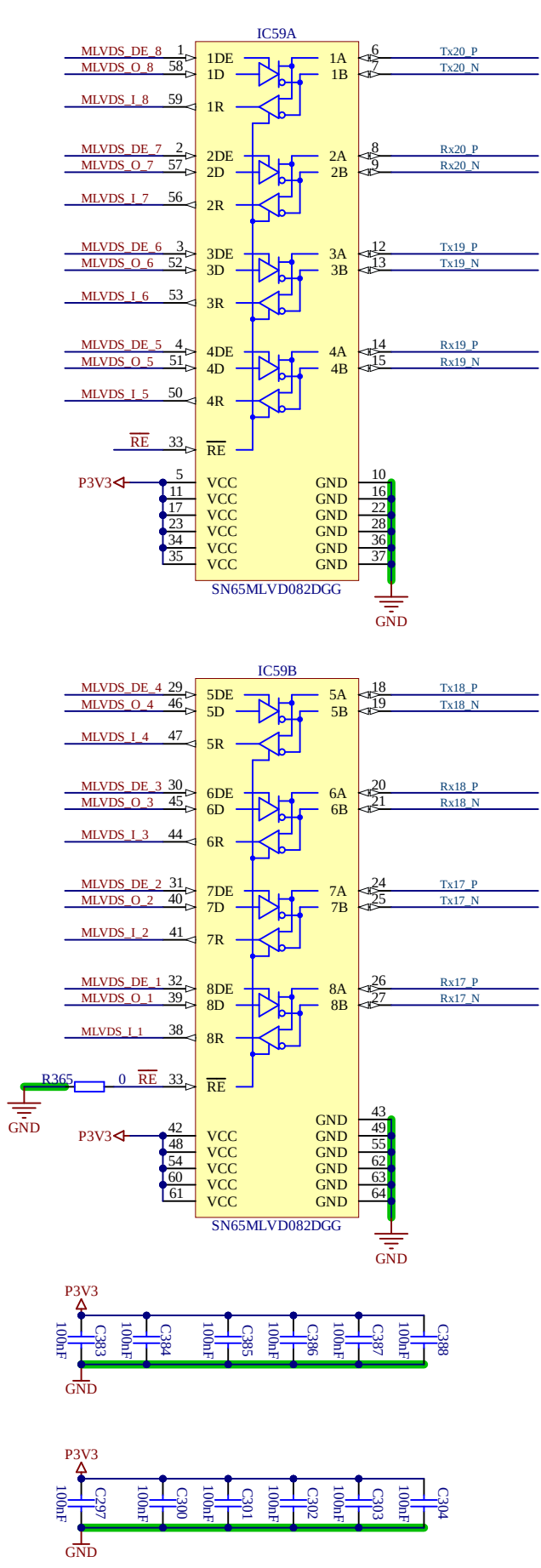
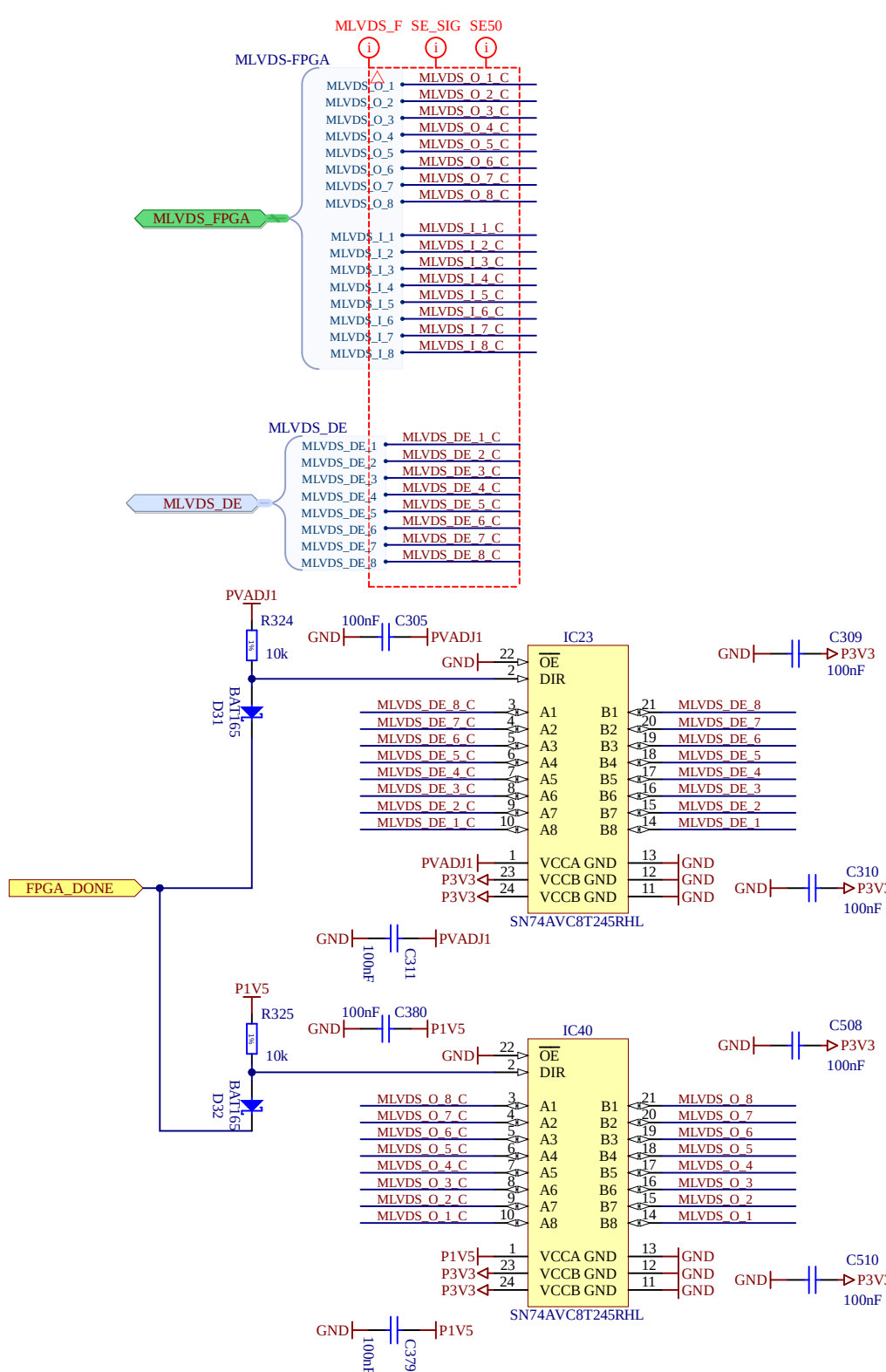
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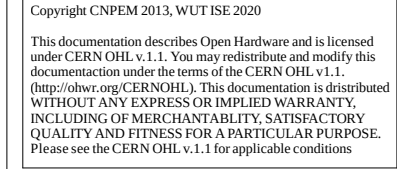
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		Designer	T. Przywozki	
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		Check by		-
		Last Mod.	F. Switkowski	19/08/2026
		File	RTM_CON.SchDoc	
		Print Date	16/09/2026 10:01:42	Sheet 20 of 29
Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE	Contact:	t.przywozki@gmail.com
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			Rev	1

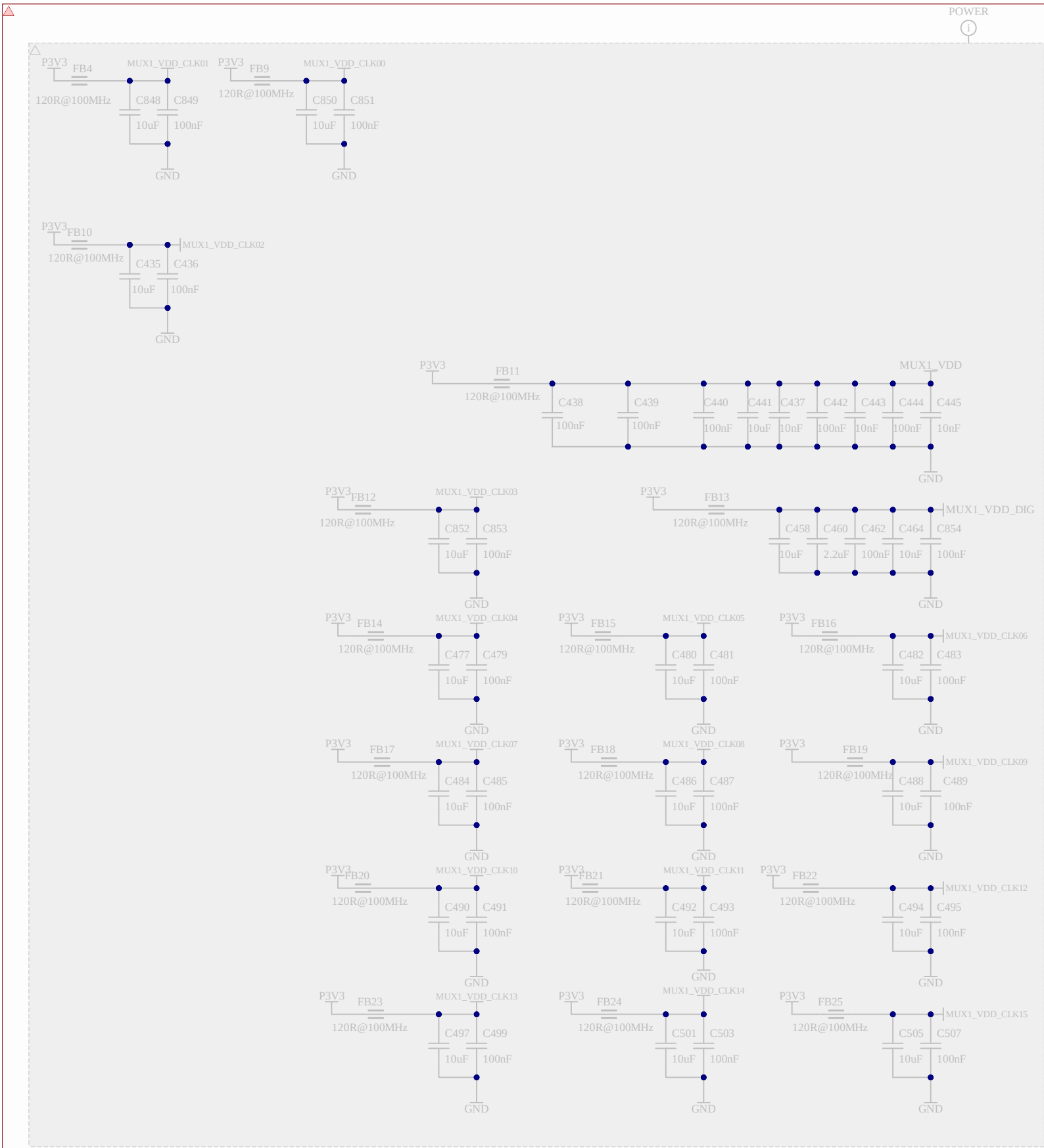
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	Document		Designer T. Przywozki Drawn by Check.by - Last Mod. F. Świtakowski 19/08/2026 File M-LVDS PHY.SchDoc Print Date 16/09/2026 10:01:43
	Warsaw University of Technology ISE		Sheet 21 of 29
	Nowowiejska 15/19 00-665 Warsaw		Size A3
	Contact: t.przywozki@gmail.com		Rev 1



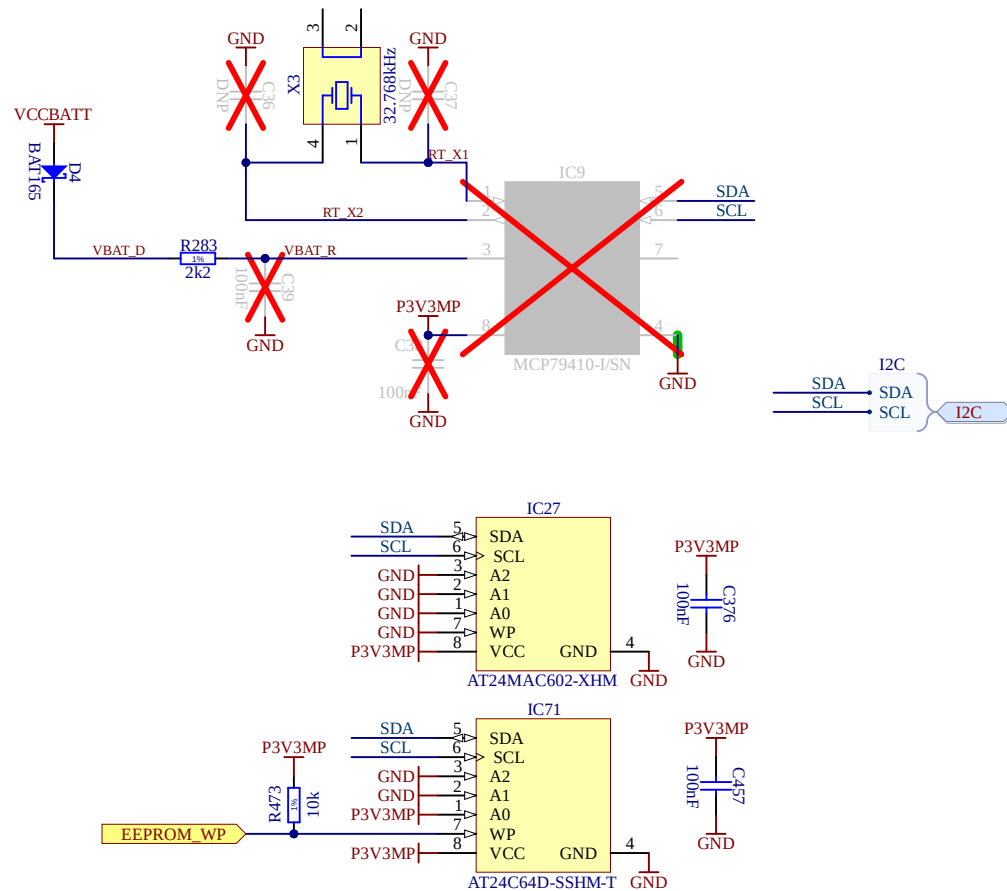




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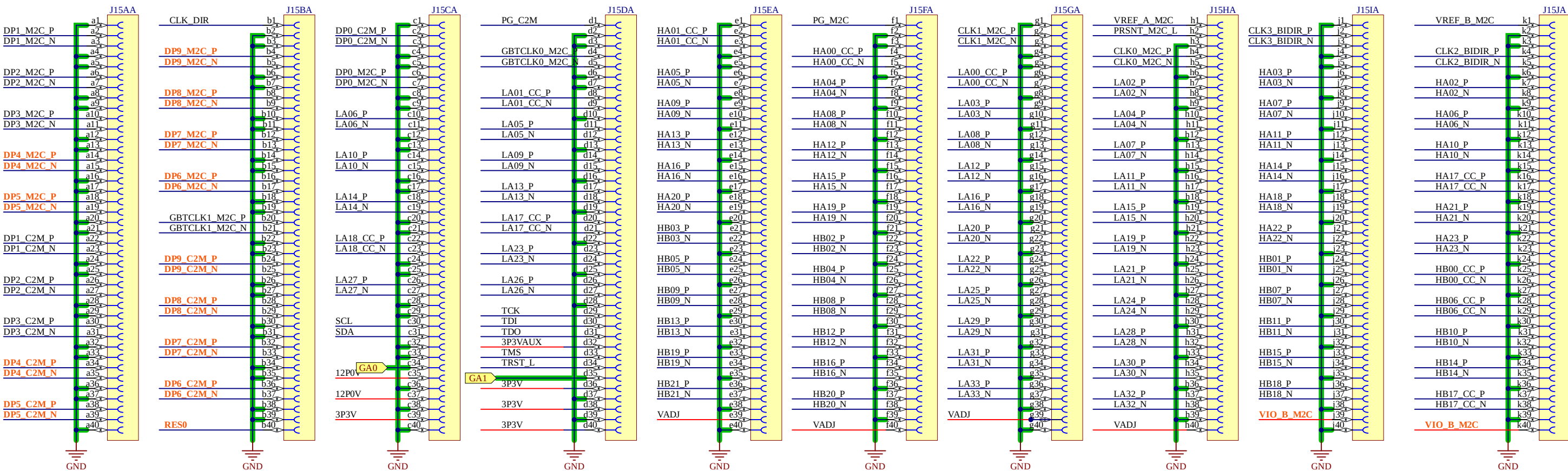
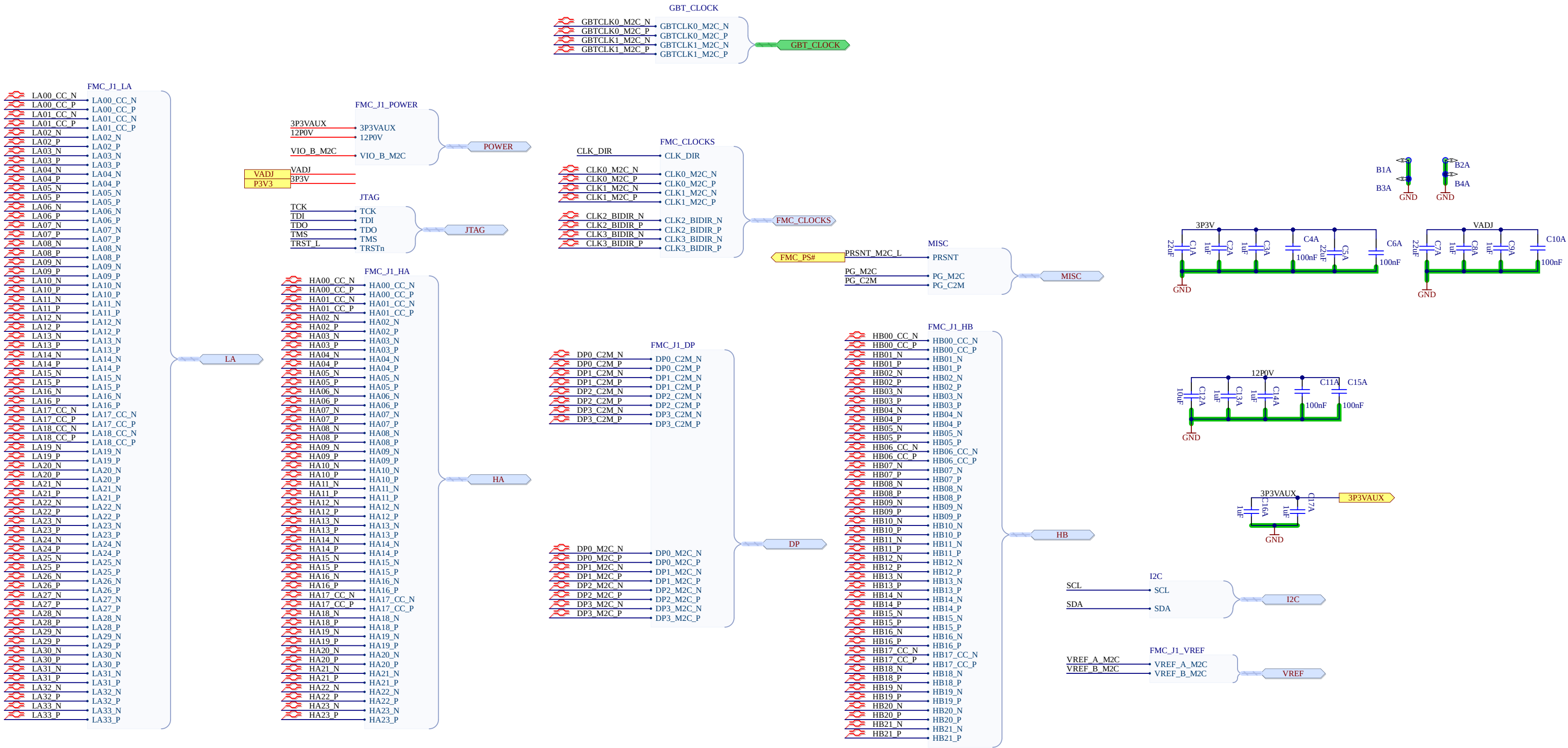
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			Check.by
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Contact: t.przywozki@gmail.com		Sheet	24 of 29
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		Rev	1



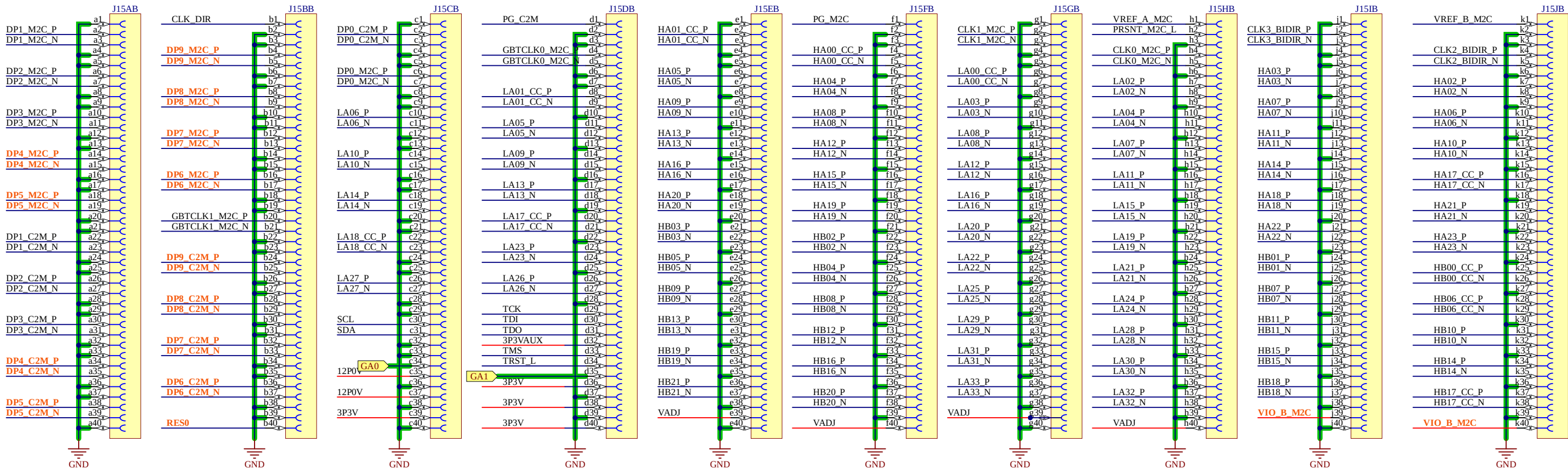
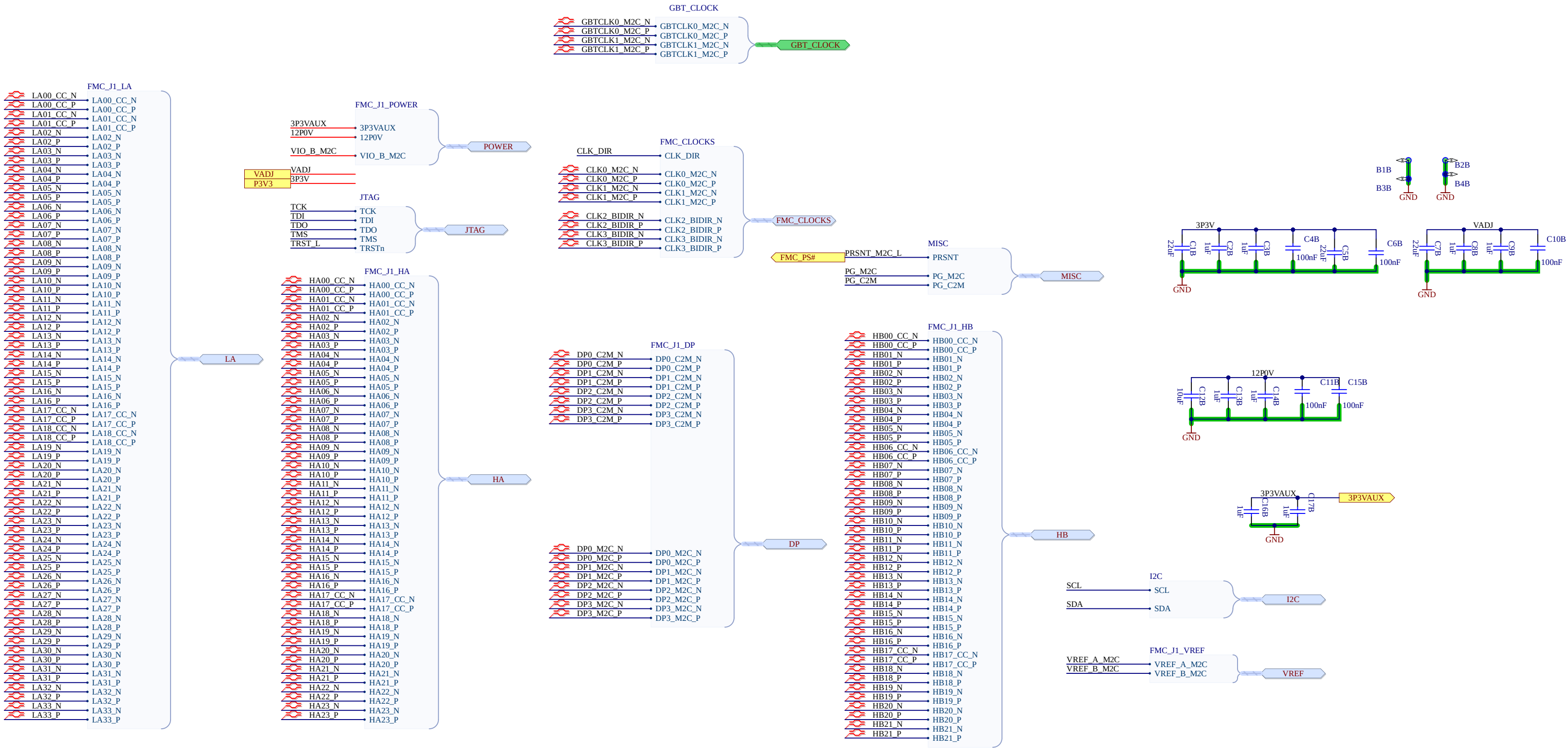
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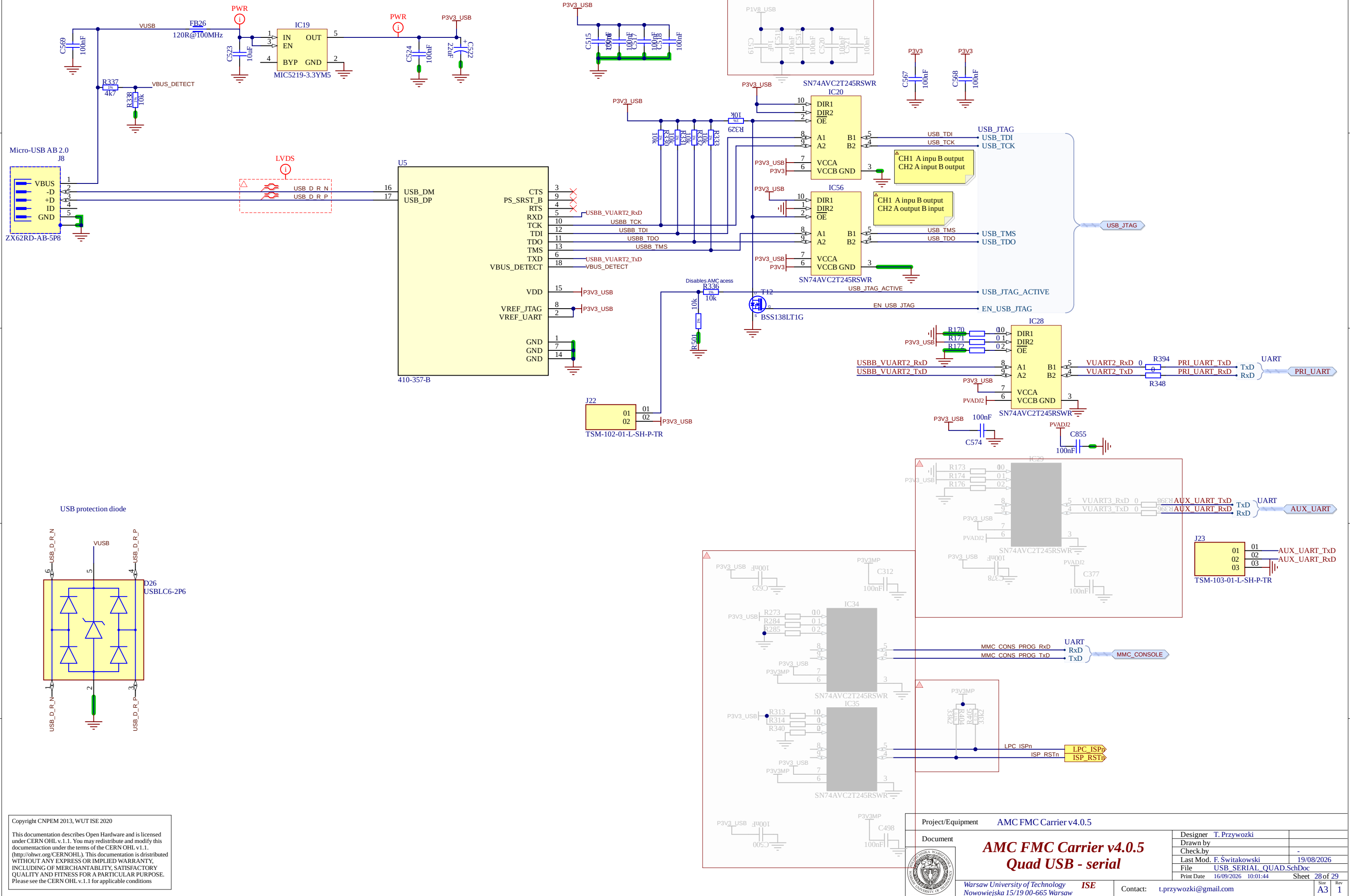
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	Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw ISE AMC FMC Carrier v4.0.5 RTCE & EEPROM		Sheet 26 of 29
	Contact:	t.przywozki@gmail.com	Size A4 Rev 1

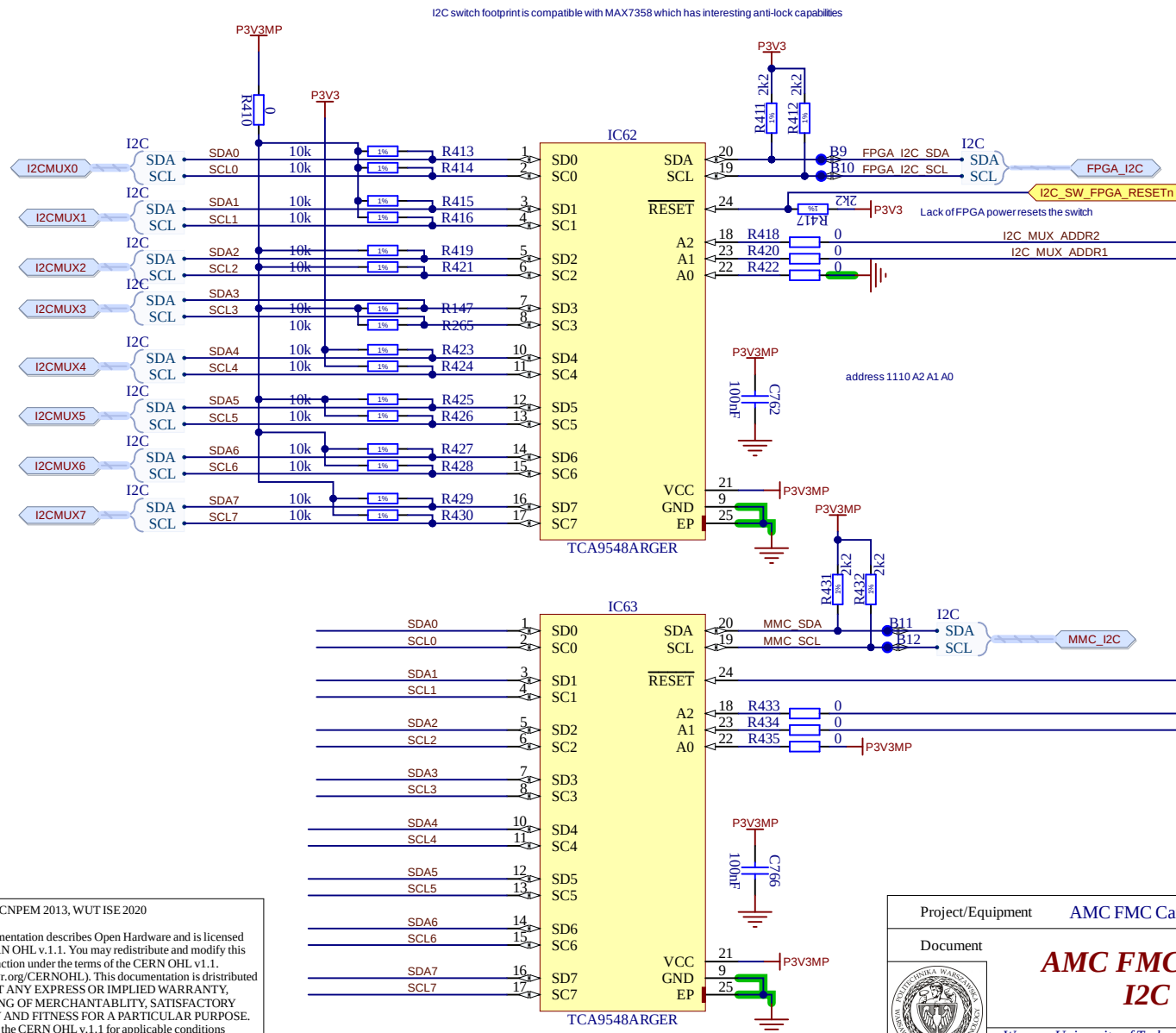


VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC



VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC





I2C device address map The resources are available from TCA9548s SW: MMC I2C1, addr 1110 A2 A1 0 FPGA I2C and MMC I2C2, addr 1110 A2 A1 1 where A2=P2[28], A1=P2[29] default addr is 0x70 and 0x71	MUX Port 0 Temp sensors	MAX6642 0x48 LM75: 0x4C LM75: 0x4D LM75: 0x4E LM75: 0x4F
	MUX Port 1 RTCE	RTC EEPROM: 0x57 SRAM/RTCC reg: 0x60 EEPROM 2k: 0x50 EU-ID: 0x59 EEPROM 64k: 0x51
	MUX Port 2	
	MUX Port 3 Power	INA3221AIRGV: 0x40 INA3221AIRGV: 0x41 INA3221AIRGV: 0x42
	MUX Port 4 Clock switch	8V54816ANLG: 0x5B
	MUX Port 5 RTM	RTM devices
	MUX Port 6 FMC2	EEPROM 0x52
	MUX Port 7 FMC1	EEPROM 0x50

mux address 1110 A2 A1 0 from FPGA side 0x7x
mux address 1110 A2 A1 1 from CPU side 0x7x

I2C MUX ADDR[2..1]

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Project/Equipment **AMC FMC Carrier v4.0.5**

Document



AMC FMC Carrier v4.0.5

I2C switches

I2C switches

Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

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Designer	T. Przywozki	
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Last Mod.	F. Świtakowski	19/08/2026
File	I2C_MUX.SchDoc	
Print Date	16/09/2026 10:01:44	Sheet 29 of 29

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Last Mod.	F. Switkowski	19/08/2026
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Size	Rev
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Rev 1
